LUT University

The multiplex of value creation and capture logics in the video game industry: An integrative review of 20 years of studies and a future research agenda

Lehtonen Miikka J., Gustafsson Robin, Hassan Lobna

This is a Post-print version of a publication

published by Elsevier

in Technological Forecasting and Social Change

DOI: 10.1016/j.techfore.2023.122756

Copyright of the original publication:

© 2023 Elsevier Inc.

Please cite the publication as follows:

Lehtonen, M. J., Gustafsson, R., & Hassan, L. (2023). The multiplex of value creation and capture logics in the video game industry: An integrative review of 20 years of studies and a future research agenda. Technological Forecasting and Social Change, 195, 122756. https://doi.org/10.1016/j.techfore.2023.122756

This is a parallel published version of an original publication. This version can differ from the original published article.

The multiplex of value creation and capture logics in the video game industry: An integrative review of 20 years of studies and a future research agenda

Abstract

With the development of digital devices, information and communication technology, and software, value creation and capture logics have evolved, and new ones have emerged. The video game industry has been a forerunner in this evolution. It has grown from an industry that revolved primarily around entertainment to one that engages game developers, users, spectators, and countless actors, becoming intertwined with many other entertainment and non-entertainment industries, such as education and health management. Although the economic importance of the video game industry has been rapidly increasing over the last two decades or so, research on value creation and capture in the video game industry has remained fragmented and predominantly based on transactional and static views. By addressing theoretical shortcomings and oversights in prior literature, this integrative review of 278 studies makes three key contributions. First, we highlight shortcomings in the current body of knowledge. Second, we craft an integrated framework of value creation and capture logics, which we have conceptualized as the multiplex of value creation logics. Third, we provide future research directions and perspectives on the future evolution of value creation and capture logics, both in and across markets and industry sectors.

Keywords: digitalization, game development, integrative literature review, value capture, value creation, video game industry

1. Introduction

The rapid evolution of digital technologies and data-intensive platforms, along with the constant advancements in computing performance, have fundamentally transformed and enhanced value creation and capture processes (Markman et al., 2021). In that regard, the video game industry has been a forerunner industry over the last decade. Firms in the video game industry utilized technological advancements and spearheaded innovations in the production and consumption of games as well as auxiliary activities around gaming, and as a consequence, new value creation and capture logics have emerged both within and outside the video game industry. In line with prior literature (e.g., Grönroos, 2008, 2011; Möller, 2006), we understand value creation logics as practices that are used to craft value propositions through combining and exchanging resources (Moran & Ghoshal, 1996). Conversely, value capture logics refer to practices that enable different actors to generate benefits from the created value (Lepak et al., 2007).

Although the economic importance of the video game industry has been increasing rapidly over the last two decades or so, research on value creation and capture in the video game industry has remained fragmented and predominantly based on transactional and static views (Wieland, Koskela-Huotari, & Vargo, 2016; Zackariasson & Wilson, 2014), where value is seen as stemming from the game, regardless of the consumer's input (Lee, 2013; Maruyama & Ohkita, 2011; Venkatraman & Lee, 2004). Such a stance seems to ignore the notion that value creation and capture in the video game industry have significantly evolved over the last twenty years especially with regards to engagement between multiple parties. For instance, rather than relying on one-time purchases, digital distribution has enabled game development companies to treat games as entry points for continuous value creation and capture and long-term customer engagement (Zackariasson & Wilson, 2013).

Furthermore, a key reason why the video game industry spearheads innovations and developments in value creation and capture is through games' unique ability to create deeply engaging and active experiences for consumers. This involvement and identification with the service consumed (e.g. characters and stories) inherently make consumers value co-creators from the get-go. Beyond gameplay, consumers utilize games as platforms for creating more value, through streaming their gameplay to others or through engaging in cosplay and other fan activities. These developments highlight the change of role for many consumers from pure consumption to a combination of consumption and production (Hart, 2017; Lehtonen et al., 2021; Pera et al., 2021).

Studies approaching value creation and capture from an engagement point of view have focused on platform-based value creation (Gawer, 2014) and network-oriented value creation (Perks et al., 2017). We argue that these two angles need to be considered simultaneously because of four unique characteristics which have become pronounced in video games production and consumption with the advancements in digital technologies. First, video games combine active player engagement with audio and visual cues in ways that other forms of entertainment cannot or are not currently doing. A player has to consider and evaluate these cues in their in-game actions, rather than passively consume them. Second, video games can either match or create demand through various value proposition configurations and offerings (e.g. new releases, expansion packs). Third, due to digital distribution, video games can be constantly customized, expanded, and turned into new services and products (e.g. fan modifications to games and spin-offs). Fourth, customers can capture different values from video games such as entertainment, skill or competence development, or social belonging. Because of these four characteristics, value creation and capture logics in the video game

industry are far from straight-forward. Due to the theoretical fragmentation of prior studies on the video game industry published in the last twenty years, in this study we set forth the following research question:

How has prior literature on the video game industry addressed value creation and capture, and what kind of theoretical shortcomings and oversights might there be?

To this end, we conducted an integrative literature review for two reasons (in line with Snyder, 2019; Torraco, 2005; Tranfield et al., 2003). First, we synthesized research on value creation and capture logics in the video game industry and illustrate some of the theoretical shortcomings and oversights in the current line of thought, and second, utilizing the critique as a springboard, we crafted an integrated theoretical framework that also formulates a future research agenda.

With this integrative literature review, we make three critical contributions to the literature on value creation and capture logics in and outside of the video game industry. First, by taking stock of how research on value creation and capture logics has evolved with respect to development and consumption of video games, we critically examine the current body of literature in order to bring to light theoretical shortcomings and conceptual ambiguities. Second, building on the critique of prior literature, we craft an integrated framework of value creation and capture logics that we have conceptualized as the multiplex of value creation logics. The multiplex specifies the various value creation and capture logics present not only in the video game industry but increasingly also in other digitized industries as well. Finally, and based on the previous contribution, we provide future research directions and

perspectives on the future evolution of value creation and capture logics more generally in and across markets and industry sectors.

2. Background: the evolution of value creation and capture in the video game industry

The first generation of commercial video games were introduced during the 1970s when the emphasis was on video game consoles and arcade games (coin-operated game machines). At the time of writing, the most recent hardware generation, the ninth, started in 2020 with both Microsoft and Sony releasing their consoles Xbox Series X/S and PlayStation 5, respectively (Gretz et al., 2019; Rietveld et al., 2019). To offer a broad overview of the industry's developments, Table 1 divides the industry into four eras based on hardware focus, dominant business models, how games are played, and the roles that have characterized both producers and consumers of video games during the eras.

Table 1. Integrative model of four eras of value creation in the video game industry.

Archetypal technological and business innovations	**Era 1: console, arcade, and computer games**	**Era 2: Commercial success of handheld game consoles**	**Era 3: Accelerated hardware development, rise of subscription-based gaming**	**Era 4: Smart / mobile devices, games as services**
Initiation	1970-	1989-	1997-	2008-
Dominant business model	Premium, games sold	Premium, games sold	Premium, subscription	Free-to-play, casual games, microtransactions

	through retailers	through retailers		
Gaming / consumption space	Home and arcades	Home and on the go	Social spaces online and offline	Seamless social spaces
Role of the provider and the consumer	Product focus and a traditional producer/consumer relationship (Bagozzi, 1975), linear value chains (Arsenault, 2017)		Servitization focus and prosumers (as per Toffler, 1980): producers as consumers of data/consumers as co-producers of value (as per Parry et al., 2012; Vargo & Lusch, 2004)	
Representative software or hardware games platforms	Nintendo Entertainment System (NES) (1983), Atari 2600 (1977)	Nintendo Gameboy (1989), Nintendo DS (2004)	Nokia's Snake (1997) and World of Warcraft (2001)	Angry Birds (2008), iPhone (2007), Nintendo Wii
Exemplary studies illustrating research foci in each era	Innovation dynamics (Gallagher & Park, 2002); network externalities (Schilling, 2003)	Revenue logics (Rajala et al., 2007); production dynamics (Tschang, 2007)	Platform dynamics (Cennamo & Santalo, 2013); virtual goods (Wang et al., 2013)	Mobile game consumption (Molinillo et al., 2020); entrepreneurship (Cabras et al., 2017)

Before digital distribution became the *de facto* way of buying and selling games, video games were predominantly sold as physical cartridges that could not be altered once they were shipped to retailers. Value was generated primarily through game developers building on existing intellectual property and hardware manufacturers creating a portfolio of games (Venkatraman & Lee, 2004). More specifically, value generation often relied on utilizing intellectual property from other forms of entertainment (e.g. movies, TV series or books) (Tang, 2017) or developing sequels for previous commercial successes (Tschang, 2007). At the same time, revenue streams relied on one-time purchases, thus emphasizing the need to focus on customer retention (Binken & Stremersch, 2009).

Initially, technological innovations made it possible to decouple software and hardware, thus creating three seminal groups of actors; hardware manufacturers, software developers, and publishers (Johns, 2006). As the literature dealing with this era of the video game industry reveals (Shankar & Bayus, 2003; Venkatraman & Lee, 2004), value creation predominantly followed the value-in-exchange paradigm (see e.g. Lusch & Vargo, 2006; Vargo & Lusch, 2004, 2008) and, especially for consumers, value proposition was usually based on the portfolio of games available for the hardware they owned or had access to. Figure 1 illustrates the archetypal value chain prior to the advent of digital distribution.

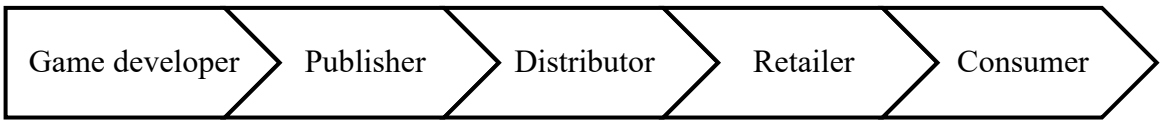


Figure 1. Value creation in the video game industry prior to digital distribution.

While initially digital distribution was thought to impact retailers first and foremost (Johns, 2006), recent years have shown a rather radical departure from what was initially a firm-

centric value creation facilitated by digital distribution (e.g. Marchand & Hennig-Thurau, 2013). Notably, consumers gained the ability to become independent creators, directly modify and expand games as well as develop their own games outside of cartridge licensing locks. Furthermore, the proliferation of mobile games and digital distribution also resulted in a surge in game development hotspots globally (such as Canada, the Nordic countries, and China) (Wilson & Zackariasson, 2012).

All of these developments have enabled multidirectional value creation and capture logics to evolve and has given rise to new business models, such as in-app purchases (Oh & Min, 2015), and subscription models (Peng et al., 2009). Thus, while initially the video game industry's value chain was portrayed in a rather linear fashion (see Johns, 2006, p. 164; Kerr, 2017, p. 84), the evolving complexity and networked nature of the industry has reached an inflection point that warrants an updated theoretical framework that explains how video games create value and for whom.

3. Methodology: integrative approach to exploring value creation and capture in the video game industry

3.1 Literature search strategy

We conducted an integrative literature review following the conventions set forth by prior literature (Elsbach & van Knippenberg, 2018, 2020; Kaur et al., 2021; Snyder, 2019; Torraco, 2005; Tranfield et al., 2003). Based on the initial assumption that prior research on the video game industry seemed to have a relatively static, product and service centered angle on value creation, the aim of the review was to scrutinize the current body of knowledge from the perspective of value creation and capture. As Torraco (2005, p. 357) writes, integrative

reviews can be used in mature topics, where the knowledge base has diversified over time, to review and criticize prior literature in order to generate novel research trajectories, concepts, or theoretical frameworks. For example, Valor et al.'s (2022) integrative review on consumers' innovation adoption criticized prior literature for ignoring the role emotions play, and consequently they proposed a framework for taking emotions into account. Similarly, Annosi et al. (2022) reviewed literature on innovations in the agricultural sector and one of their main critiques of the literature was the apparent absence of individuals when making theoretical claims. To counter this, they (ibid.) utilized prior literature as a springboard to craft a framework that balances structures and individual agency. Thus, integrative reviews – when dealing with mature topics – can be understood as serving two functions: first, review prior literature to expose conceptual or theoretical oversights, and second, generate new theoretical frameworks, vantage points, and / or avenues for moving forward (Torraco, 2005, p. 363).

To craft our literature search strategy, we followed conventions from prior integrative reviews (e.g. Annosi et al., 2022; Torraco, 2005; Valor et al., 2022; Whittle et al., 2023). More specifically, our strategy consisted of four stages (Figure 2).

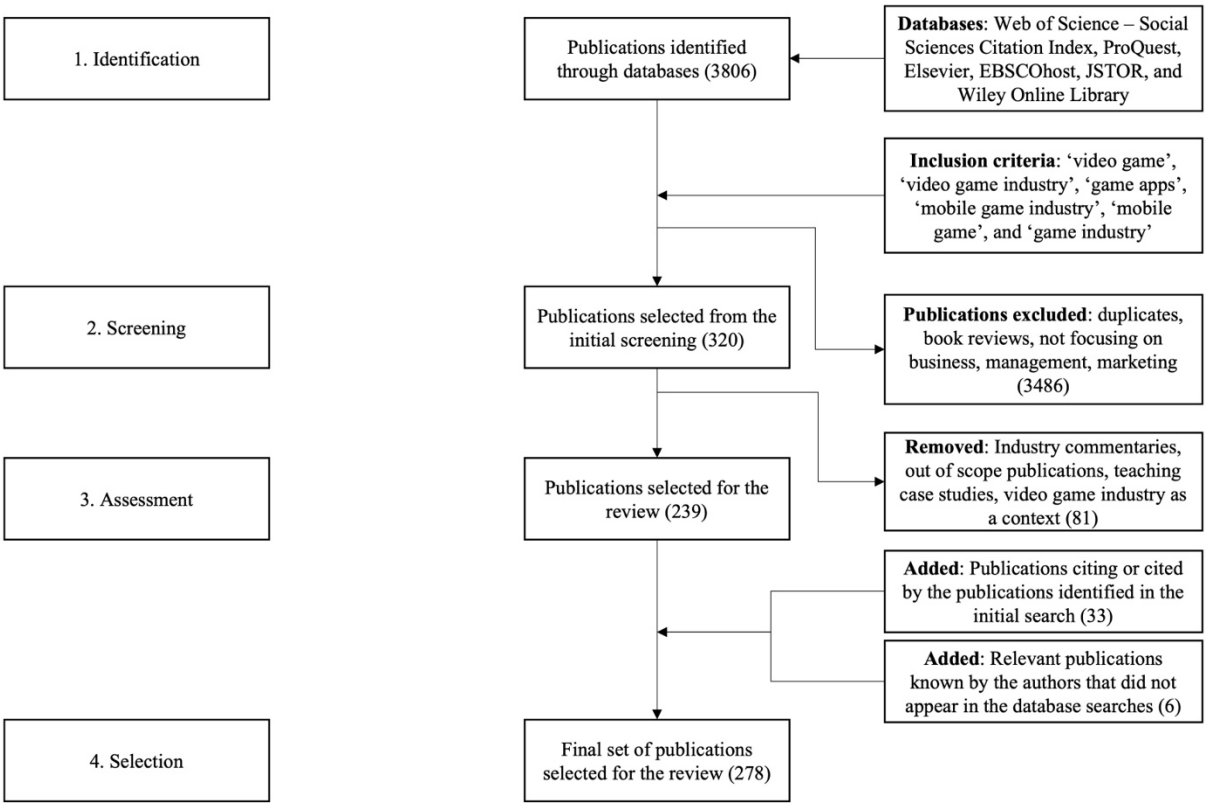


Figure 2. Literature search strategy.

First, we laid out the foundations for the search by defining the conceptual search boundaries. Acknowledging the diversity of keywords utilized in prior literature on the video game industry (Abbasi & Jamak, 2017; Alpert 2007; Quandt et al., 2015), we searched for relevant literature by using the following keywords: 'video game', 'video game industry', 'game apps', 'mobile game industry', 'mobile game', and 'game industry'. In terms of the databases, we covered Web of Science – Social Sciences Citation Index, ProQuest, Elsevier, EBSCOhost, JSTOR, and Wiley Online Library to ensure we would cover as much literature as possible. Although Torraco (2005, p. 360) suggests using Web of Science as the point of departure, we opted for a broader search given that Web of Science alone does not necessarily cover all the relevant literature. While we did not limit the starting year of the search, no studies falling under the scope of this review were identified prior to 2001. As

such, by focusing on peer reviewed studies published in English between 2001 and 2021, our initial search yielded 3806 entries.

In the second step, we removed book reviews, duplicates, and studies that did not focus on business, marketing, and management. Especially the last exclusion criterion is based on the reasoning that value creation and capture are topics essentially covered in the three aforementioned fields. At this stage, we checked the entries' title and abstract (as per Annosi et al., 2022) to compare them against the exclusion criteria. The first author did the initial screening, after which both second and third author double checked the screening to ensure consistency. As a result, 3486 entries were removed, thus leaving us with 320 entries.

The third step focused on us going through the 320 entries in more detail. Additional 81 publications were removed since they were industry commentaries, case studies for teaching purposes, studies using the video game industry as a context for studying something else (i.e., studies not contributing to our understanding of the game industry itself), or studies looking at wild game (e.g. hunting elephants and antelopes). Once again, the first author went through the initial data set, and afterwards we collectively ensured that the remaining titles adhered to our initial inclusion criteria. Thus, the initial screening yielded 239 entries in total. Noteworthy is that the coding and analysis process was conducted by the lead author of this research. As such, validation of the coding was not possible (e.g. through checking agreement rates between coders). This single-author coding can mean a relatively high internal consistency in coding, as coding was conducted according to a singular mindset, but it also means that this review is subject to researcher bias and human error as is communicated in the limitation section of this study.

Finally, we complemented our search in two ways: first, we adopted a snowballing technique to identify additional publications from the 239 entries (backwards and forwards searches) (e.g. Peltoniemi, 2015; Ravasi & Stigliani 2012), and second, we included sources that we had identified as relevant but had not appeared during the database searches (as per Whittle et al., 2023). Each author did this individually, after which we presented our suggested entries to each other to ensure they met the inclusion criteria. This part of the search helped us identify 39 additional publications, thus bringing the dataset to 278 publications (see Appendix 1 and 2 for more details regarding the data set). To ensure all 278 entries met our inclusion criteria, we went through their title and abstract once more.

3.2 Data analysis

In lieu of standardized data analysis processes in integrative reviews (Snyder, 2019; Torraco, 2005), we mobilized the Gioia methodology (Gioia et al., 2013) that is well suited for rigorously developing new concepts and theoretical frameworks. In essence, the Gioia methodology consists of three interrelated stages (Gioia et al., 2013, p. 20) – 1st order categories, 2nd order themes, and aggregate dimensions – that begin with a strict focus on the empirical data and end with developing theoretical insights from the data. In addition, we also took inspiration from prior integrative reviews (Annosi et al., 2022; Valor et al., 2022; Whittle et al., 2023) in terms of how they structured their data analysis process. This was done to ensure rigor and transparency in how we analyzed the data. Figure 3 visualizes how we derived insights from the data set.

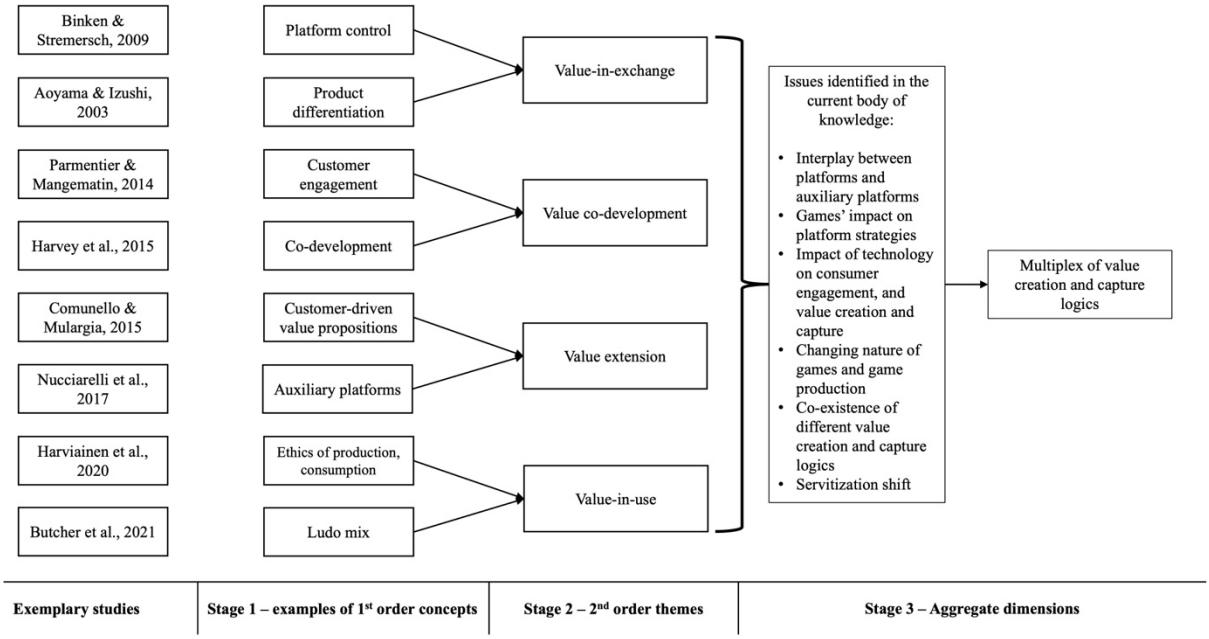


Figure 3. Data analysis process visualized.

In the first stage, we screened each publication from a value creation and capture perspective; namely, who creates and / or captures value, through what logics, and with what outcomes. While the first author created the initial data coding, second and third author served as sort of devil's advocates: their role was "to critique interpretations that might look a little too gullible" (Gioia et al., 2013, p. 19). We then jointly discussed the analysis and went through the publications again to ensure we collectively agreed on the coding.

Second, based on the first stage, we started categorizing the publications based on their underlying assumptions about value creation and capture. Here, as suggested by Gioia et al. (2013), empirical data is contrasted with prior literature to see whether existing concepts and frameworks could help us explain what the data is telling us. As we started forming categories, we realized that mobilizing pre-existing concepts helped us distinguish and characterize four distinct value creation and capture logics. *Value-in-exchange* (e.g. Schilling, 2003) describes value as exclusively being created between developers, publishers, and other

firms so that games inherently contain value regardless of consumer response to it. *Value co-development* (Haefliger et al., 2010), on the other hand, involves the consumers as creators in the game development process from the get-go, thus highlighting the co-creational dimension of value. *Value extension* (e.g. Prahalad & Ramaswamy, 2004; Roberts et al., 2015) highlights the role of consumers as innovators and value co-creators through game companies partially opening up their intellectual property to them. Here, consumers are not modifying the existing game, but instead building new versions of it or creating content for it. Finally, *value-in-use* (e.g. Hsiao & Chen, 2016) argues that games have value based on the interaction between the game and the consumer, i.e., how the game is interpreted, made sense of, and actively used in the life of a consumer.

Finally, in the third stage, we exposed the four value creation and capture logics into critical scrutiny (as suggested by Torraco, 2005). We focused on delineating key theoretical contributions in prior literature in order to identify "any deficiencies, omissions, inaccuracies, and other problematic aspects of the literature" (ibid., p. 362). While we recognize that categorizing prior literature from the perspective of value creation and capture is but one way of synthesizing the literature, at the same time this vantage point has yielded two crucial theoretical insights. First, prior literature seems to have predominantly ignored the impact technological changes and innovations have had on the video game industry's value creation and capture logics, thus resulting in fragmented theorization efforts. Second, building on the previous point, these logics are not mutually exclusive but through co-existence they influence all levels of the video game industry (micro – meso – macro). Thus, in the following section we cover the four value creation and capture logics, after which we present the integrated theoretical framework as well as the future research agenda.

4. The multiplex of value creation logics in the game industry

In this section we focus on separately synthesizing the four distinct value creation and capture logics in the video game industry. To begin with, Table 2 offers an overview of the four value creation and capture logics by focusing on conceptual underpinnings as well as typical methods and foci.

Table 2. Theoretical and conceptual overview of the four value creation and capture logics.

Value creation and capture logic	Conceptualizations of value creation and capture	Typical methods / data	Typical topics	Example studies
Value-in-exchange	Value creation as coordination between developers, publishers, and hardware manufacturers; value capture is based on consumer lock-in effects	Historical; archival data	Historical and cultural factors that explain the emergence of the video game industry in a specific regional context	Aoyama & Izushi, 2003;
	Companies capture value from product differentiation; technological changes influence the strength of product differentiation	Historical approach; archival data; interviews	How firms capture value from creating differentiated game intellectual property	Ernkvist & Ström, 2018;
	Companies capture value from creating games focusing on longevity	Historical data; quantitative; consumer behavior surveys	What factors influence consumers' time spent on video games	Kaimann et al., 2018;
	Value capture from network effects; value creation based on how many games and complementary goods a hardware has	Historical; industry-level quantitative data	How companies utilize technology to challenge incumbents or to prevent competition	Schilling, 2003;
	Value creation emerges from companies differentiating their offer from each other; value capture is based on consumer lock-in effects	Historical; secondary data; literature review	How industry-level paradigm shifts influence competition between companies	Zackariasson & Wilson, 2010;
Value co-development	Consumers engage in value creation through co-creating innovations; value capture	Multiple case study; interviews; firm-level analysis	How users become entrepreneurs through community engagement	Haefliger et al., 2010;

	emerges from consumers becoming entrepreneurs to monetize the innovations			
	Consumers create new ideas for games (value creation) and companies develop them (value capture)	Explorative case study; discussion forums; interviewing consumers and producers	How companies and user communities learn and develop their knowledge base	Jeppesen & Molin, 2010;
	Game designers as creating value through unorthodox game ideas; consumers providing financial backing to such projects	Discourse analysis; Kickstarter crowdfunding projects	How crowdfunding reshapes consumer and producer roles	José Planells, 2017;
	Online user communities as value creators; producers capturing value through engagement and consumers from better products / services	Multiple case study; interviewing consumers and producers; secondary data	How consumers and producers of video games engage in co-development	Parmentier & Mangematin, 2014;
	Value creation as a creative process; value is captured from consumers' perceptions of novelty and traditions	Archival data; practitioner interviews; ethnographic observations	What kind of strategies firms develop to balance between novelty and traditions	Tschang, 2007;
Value extension	Firms create value by partially opening up their intellectual property; consumers capture value from modifying the IP; companies capture value from such modifications	Economic analysis	How firms utilize consumer networks to create new value propositions (i.e. value creation)	Arakji & Lang, 2007;
	Firms capture value from engaged consumer communities; value created simultaneously by	Secondary data; interviews; case study	Identifying consumer typologies from the perspective of value extension	Burger-Helmchen & Cohendet, 2011;

	communities and firms in digital environments			
	Companies capture value from what consumer perceive they can do with the game; consumer value capture through fulfilling hedonistic needs	Conceptual; interviews;	Consumer behavior in digital environments	Denegri-Knott & Molesworth, 2010;
	Firms create value through crowdfunding by unifying channels that bring resources; consumer create value from collaborating with the firm; value capture emerges from resource coordination	Multiple case study; practitioner interviews	How crowdfunding influences and transforms value creation processes in the video game industry	Nucciarelli et al., 2017;
	Consumers capture value based on their motivations for engaging in co-creation efforts; firms capture value from managing their relationships with consumers	Qualitative; practitioner interviews	Why consumers engage in co-creation with game development companies	Roberts et al., 2014
Value in use	Firms capture value based on consumers' perceptions of product quality; video games create value for consumers based on internal and external preferences	Video game sales figures; archival data	How reviews of video games influence consumers' purchase intentions	Cox & Kaimann, 2015;
	Firms create value through the free-to-play business model and capture value from balancing consumer enjoyment	Quantitative; consumer survey	How firms balance enjoyment and monetization	Hamari, 2015;

	Consumers' value perception of free-to-play mobile games; producers monetize (capture value) consumer loyalty	Quantitative; consumer survey	How consumer perceived value and loyalty are interconnected	Molinillo et al., 2020;
	Consumers perceive value differently based on their needs and expectations; firms capture value from high quality and well-functioning entry to the game	Exploratory study; quantitative; consumer survey	What kind of needs and values consumers have related to mobile games	Penttinen et al., 2010;
	Consumers perceive value in nostalgia, storytelling; producers monetize consumers' desire for nostalgia	Mixed methods; interviews, consumer survey	How producers capture value from combining value propositions	Tang, 2017;

4.1 Value-in-exchange: When producers and platforms dominate value creation and capture

Until the end of the 20th century, the video game industry was dominated by three major game console manufacturers and platform providers (Sony and Nintendo in Japan, Microsoft in USA). Competition between these few actors attracted a notable number of studies (Cenamor et al., 2013; Cennamo & Santalo, 2013; Huotari et al., 2017; Maruyama & Ohkita, 2011; Prieger & Hu, 2012; Zhu & Iansiti, 2012). At the same time, Nintendo's decision to control third-party developers more strictly through licensing fees and lockout chips quickly became an industry standard that, to a large extent, is still prevalent in the 21st century (Chatfield, 2011). Thus, to prevent value destruction and cannibalization of the market through rampant releasing of cheap and/or low-quality hardware and software, value creation and capture in this context became predominantly focused on ensuring product quality by controlling which game development companies were granted development licenses.

Hence, research falling under this category has highlighted relatively significant homogeneity in terms of the actors involved in value creation and capture, and this has been shown to stem, to a large extent, from non-compatibility strategies (e.g. Cennamo & Santalo, 2013; Zhu & Iansiti, 2012). Since games were sold on physical cartridges, value capture predominantly focused on one-time purchases and lock-in logics through games not being made compatible across hardware (Schilling, 2003; Situmeang et al., 2013). Through control, hardware manufacturers achieved two goals: first, allocation of intra-firm resources between game development companies (quality aspect), and second, better portfolio management by ensuring that the games sold for consoles are complementary rather than cannibalizing (Binken & Stremersch, 2009; Shankar & Bayus, 2003). As Figure 4 illustrates, hardware manufacturers (platform providers) and game development companies (asset providers)

created value propositions to the consumers, and value capture emerged from locking in customers to a specific platform by focusing on quality and a complementary product portfolio (e.g. Gretz, 2010b).

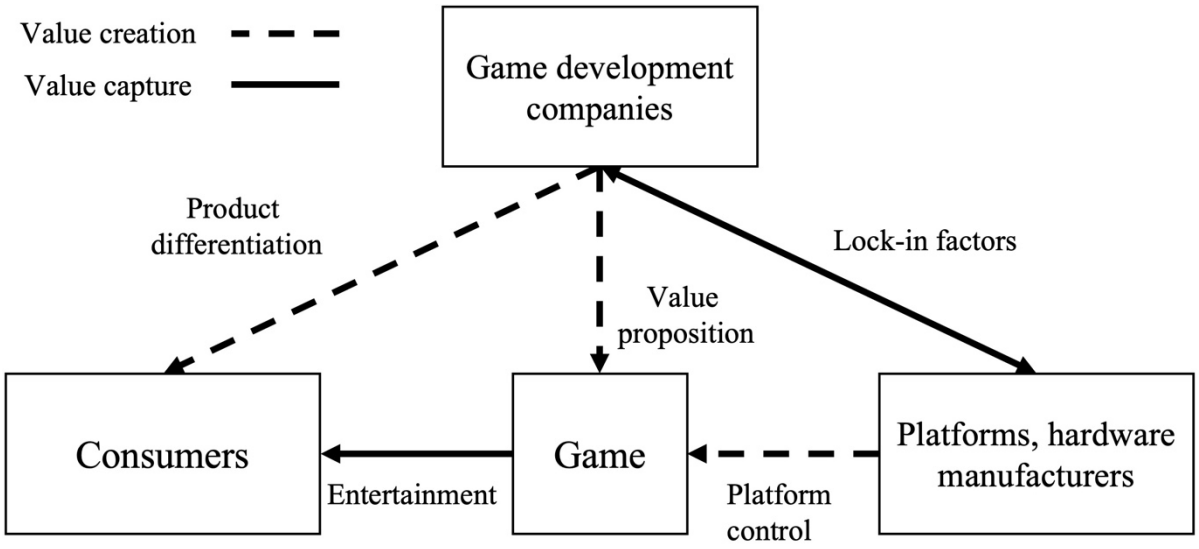


Figure 4. Value-in exchange logics in the multiplex of value creation and capture in the video games industry.

The continued duopolistic nature of the video game console market (Ewalt, 2008; Subramanian et al., 2011) frames the actions of game development companies and hardware manufacturers. More specifically, studies focusing on networks and network effects (Anderson et al., 2013; Balland et al., 2013; Gretz & Basuroy, 2013; Inoue & Tsujimoto, 2018; McIntyre & Srinivasan, 2017; Shankar & Bayus, 2003; Sun et al., 2004; Venkatraman & Lee, 2004) and product and pricing strategies (Ebina & Shimizu, 2009; Gretz, 2010a; Liou & Gao, 2011; Liu 2010; Nair, 2007; Readman & Grantham, 2006; Seidl et al., 2017) have illustrated how value is created through platform control. For example, hardware manufacturers often rely on exclusive titles (games released on one only platform such as the *Halo* franchise for Microsoft's Xbox and the *Uncharted* series for Sony's PlayStation) to

differentiate themselves from competitors. Both Microsoft and Sony have often acquired stakes in companies developing these exclusive titles to further increase the barriers of these developers' entry to the hardware of the other competitor. In other words, when the consumers have fewer options for hardware (game consoles) due to the incompatibility of games across hardware, firms can focus on value capture through inter- and intraorganizational cost optimization.

Historically a large proportion of the game development companies and hardware manufacturers that came to dominate the video games markets globally were based in Japan and the US (Kerr, 2006, 2017). They developed geographical proximity advantages, such as decreased costs in network coordination as well as knowledge transfer (e.g. Aoyama & Izushi, 2003; Izushi & Aoyama, 2006), which they have maintained quite successfully over time. Consequently, most industry-level studies have focused on these two countries (Aoyama & Izushi, 2003; Clements & Ohashi, 2005; Consalvo, 2006; Cox, 2006; Izushi & Aoyama, 2006; Schilling, 2003; Zhou, 2017), with a few notable exceptions looking at other regions (Cao & Downing, 2008; Darchen & Tremblay, 2015; Wang et al., 2016). This relative homogeneity is also extended to game development processes (Johnson, 2013; Kang & Kim, 2015; Parker et al., 2017; Takeaki, 2011; van den Ende et al., 2013; Walfisz et al., 2006), small and medium-sized enterprises (Cunningham et al., 2012; Dixon & Karboulonis, 2001; Gandia & Gardet, 2019; Hotho & Champion, 2011; Martin & Deuze, 2009), and business models and value creation (Davidovici-Nora, 2014; Huang, 2016; Rayna & Striukova, 2014; Waldner et al., 2013). In essence, value creation and capture strategies and logics are often restricted, based on the combined effects of technological limitations and geographical collocation of firms.

4.2 Value co-development: The pre-launch perspective

Value co-development involves consumers in the game development process as creators prior to the launch, thus highlighting the co-creative value of games and gaming (Haefliger et al., 2010). Creators contribute to the game development company's initial value proposition (e.g. through playtesting or providing ideas for the game) (Koch & Artmayr, 2020) and, by the same token, company captures value from consumer engagement. For instance, consumers can provide feedback to the company (Jeppesen & Molin, 2003), serve as brand ambassadors (Parmentier & Mangematin, 2014), or provide initial capital in the form of crowdsourcing (Smith, 2015). Consumers can also be seen as critics, creating new value through offering alternative pathways forward by pointing out shortcomings in the existing value proposition (Smith, 2015). Consumers can simultaneously be critics and creators, and similarly these roles are not static, but instead fluctuate over time and across games. While one stream in the reviewed literature has extensively looked at the benefits and perils of co-development from a firm's point of view (Arakji & Lang, 2007), another stream explored the consumer perspective as creators and critics participating in developing or transforming the value proposition of the game developer (Copplestone, 2017).

From a company's perspective, involving the consumers can therefore validate or reject the initial value proposition. From the consumer's perspective, value creation has the potential to move from transactions to relationships, and subsequent value capture strategies have diversified. For instance, pre-release involvement can be used to manifest commitment or receive exclusive in-game content. However, from a platform's perspective the evolving relationship between consumers and game development companies is akin to a double-edged sword: on the one hand, it can strengthen the lock-in effect to a platform where the consumer-developer relationship is taking place (Koch & Artmayr, 2020), while on the other hand, the

relationship might evolve away from a platform, as consumers can opt to switch platforms to follow the game development company which they initially started with (Planells, 2017).

Co-developing with customers or through open innovation practices has received considerable research attention, including Greer and Lei's (2012) seminal literature review across several industries. Co-development has both enabling (e.g. facilitating customization) and hindering factors (time pressure limiting the potential gains). While Greer and Lei (2012) reviewed papers focusing on the video game industry, our review expands on that work, showing how the video games industry extends our current understanding of consumer co-development in at least two ways. First, since game development processes often become public after a game has been launched, consumers can create value for the company by becoming brand advocates. This, however, does not necessarily imply the company is bound to benefit from such consumer commitment: failure to launch the game can negatively impact the company's value creation and capture processes as said brand advocates might turn against the company. Second, and building on the last point, by showcasing gameplay of a game in development (e.g. on YouTube or Twitch), consumers can involve other consumers for additional benefits to themselves and the developer in terms of feedback, awareness, and streaming fame. In this case, consumers can capture value in the form of status and recognition, while companies can capture value from increased consumer engagement. For example, Unknown Worlds Entertainment (developer of *Subnautica*), opened up the game development process to consumers and streamers reportedly for two main reasons: to gather early customer feedback as well as generating hype prior to release (Bötel, 2019). Consequently, the game development process prior to launch has now become more transparent, especially towards consumers (Figure 5) (Banks & Potts, 2010).

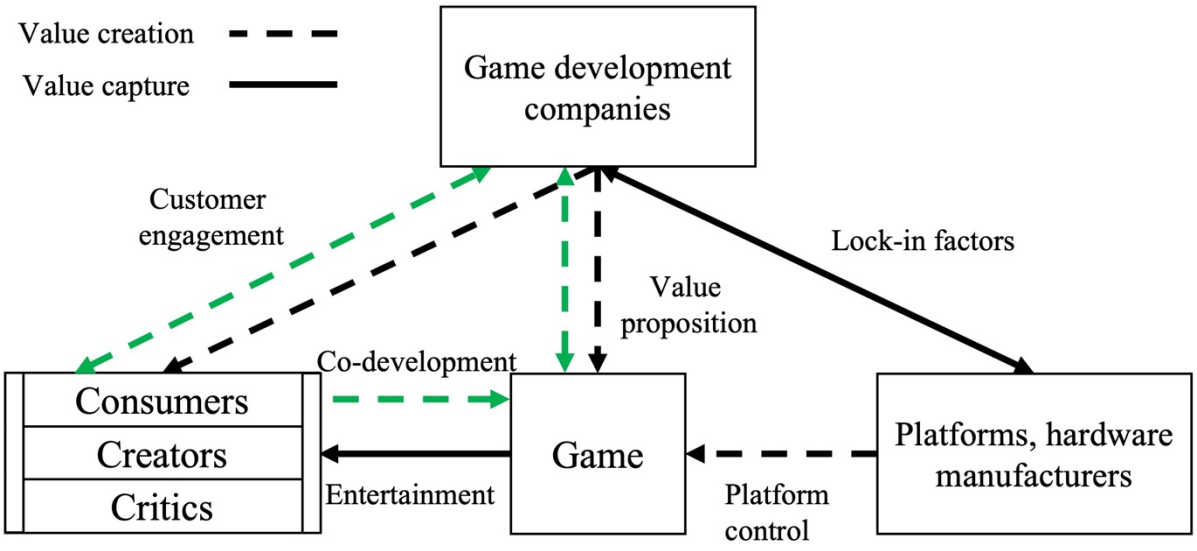


Figure 5. Value co-development logics in the multiplex of value creation and capture in the video games industry.

As value creation and capture logics in the video game industry have diversified, partially due to co-development, managerial and operational practices in game companies have also been impacted. From a value creation and capture perspective, increased competition for consumers and their resources have pushed companies to explore new business models. In other words, one-time purchases have given way to other emergent business models such as free-to-play (Kerr, 2017; Cennamo & Santalo, 2019), and this transformation has trickled down to the employee-level in the form of having to acquire new skills or be pushed away from the industry (Chung & Kwon, 2020). Competition for professionals has intensified, and at the same time work has become more precarious (Bulut, 2015). Some studies have focused on talent management and retention (Cadin & Guérin, 2006; Christina, 2016; Scholz, 2012; Teipen, 2008), and by the same token the fragility of project-based work has also received scholarly attention (Bulut, 2015; Christina, 2016; Hodgson & Briand, 2013; Peticca-Harris et al., 2015; Wright, 2015).

These two perspectives seem somewhat contradictory, which is partly explained by how video games often struggle with balancing artistic expression and commercial gains (Autier & Picq, 2005; Broekhuizen et al., 2013; Tschang, 2007). On the one hand, being employed in the video game industry has often been seen as a transformation from a hobby to a career (Bulut, 2015; Haefliger et al., 2010), but on the other hand, the passion and commitment of gamers has often been exploited by game development companies, especially by those developing so called AAA games (games with the highest production and marketing budgets) (Schumacher, 2006). Thus, while the value co-development paradigm comes with several firm-centric benefits centered on marketing, consumer engagement, and diversified revenue streams, as a consequence the producer–consumer balance has also become somewhat skewed through value creation and capture logics converting consumers' enjoyment into labor (see e.g. Goggin, 2018).

4.3 Value extension: The post-launch perspective

Value extension refers to consumers' involvement in value creation processes taking place after a game has been developed (Roberts et al., 2015). While value-in-exchange emphasized value creation emerging from the relationship between developer and hardware manufacturer and value co-development drew attention to the processes and practices *prior* to game release, value extension focuses on the type of consumer involvement that does not necessarily involve the producer anymore, specifically, relations with the platform on which a game is published and made available to consumers. Because contemporary platforms allow games to be updated, or even modified by players, platforms have come to treat games as services rather than products (Inoue & Tsujimoto, 2017). Consequently, game value proposition has the potential to change (expand, shrink, or branch out) after a game has been launched.

Such technological liberation also impacts market dynamics by creating more competition between game developers (Gretz et al., 2019). This is because value creation logics and strategies tend to favor platforms, as their revenue is based on the aggregate revenue of the developers they publish, pushing them to release more games for more developers. These same mechanics that benefit the publisher, create more competition for developers, risking their revenues and longevity. In addition, this liberation lends itself also to companies being able to utilize diverse business models (i.e. premium, free-to-play, or hybrid games) for creating value due to the notion that each type requires somewhat specific skills and competences. For example, free-to-play games rely heavily on data analytics to investigate consumers behavior patterns, whereas premium games focus more on developing distinct intellectual property (Tschang, 2007).

Consumers have also appropriated these same value creation and capture liberation logics for their own purposes, independent from a game development company's strategic objectives. For example, the mobile game *Clash of Clans* by Supercell encourages people to form groups within the game. Eventually, according to the company, some players have used the chat function in their group to ask for help in matters not related to the game (Lehtonen & Harviainen, 2016). Other studies (Zimmerman, 2019) have found similar ways through which consumers branch out from a developer's intended value proposition to create, and ultimately capture, value that revolves around social interaction and community-building. At the same time, Arakji and Lang (2007) showed how Valve was able to stimulate consumer value co-creation and, eventually, value capture through increased consumer engagement. Valve opened the engine of their game *Half-Life*'s for consumers to create modifications based on the original game. The business relevance here is succinctly illustrated by Burger-Helmchen

and Cohendet (2011, p. 318) as follows: “One of the most striking features of the video game industry is that a significant part of the value is created by cognitive resources (the communities), which are not directly controlled by the firm”. Thus, value creation and capture are no longer determined by the firm, but consumers are extending existing value propositions either within predetermined parameters or by appropriating the game to serve other purposes such as social interaction or status.

Whereas the discussion above revolved around value creation and capture emerging from consumers and the game development company interacting with each other, players creating content on streaming services such as YouTube and Twitch showcases how players can create value without a company’s direct involvement (Törhönen et al., 2020, 2021). Decisions to design a game that is suitable for streaming are usually made early on in the development process (Johnson & Woodcock, 2019), and the reasoning behind allowing it from a developer’s perspective is two-fold: first, the free-to-play business model aims at capturing value through long-term and continuous monetization (Cabeza-Ramírez et al., 2021). Hence, to avoid consumer attrition, streaming has been a means to balance value creation and capture (Xu et al., 2021); and second, streamers can engage with the game in hitherto unknown ways, thus serving, not only as game champions, but also as idea and insight providers to games companies (Sjöblom et al., 2019). For example, games like *League of Legends*, *Brawl Stars*, and *Dota 2* have been designed so as to promote streaming through live spectating, tournaments, and multiplayer modes. In a somewhat different example, streaming chess seems to have increased in popularity especially during the Covid-19 pandemic (Browning, 2020) potentially as an avenue for human connection and interaction. Here, since no firm is directly monetizing chess as a game, apart from selling chess sets and accessories, we can see how consumers as players are monetizing their

gameplay and extending the value offering of chess as a game. Figure 6 shows how value creation and capture practices have diversified even further through value extension.

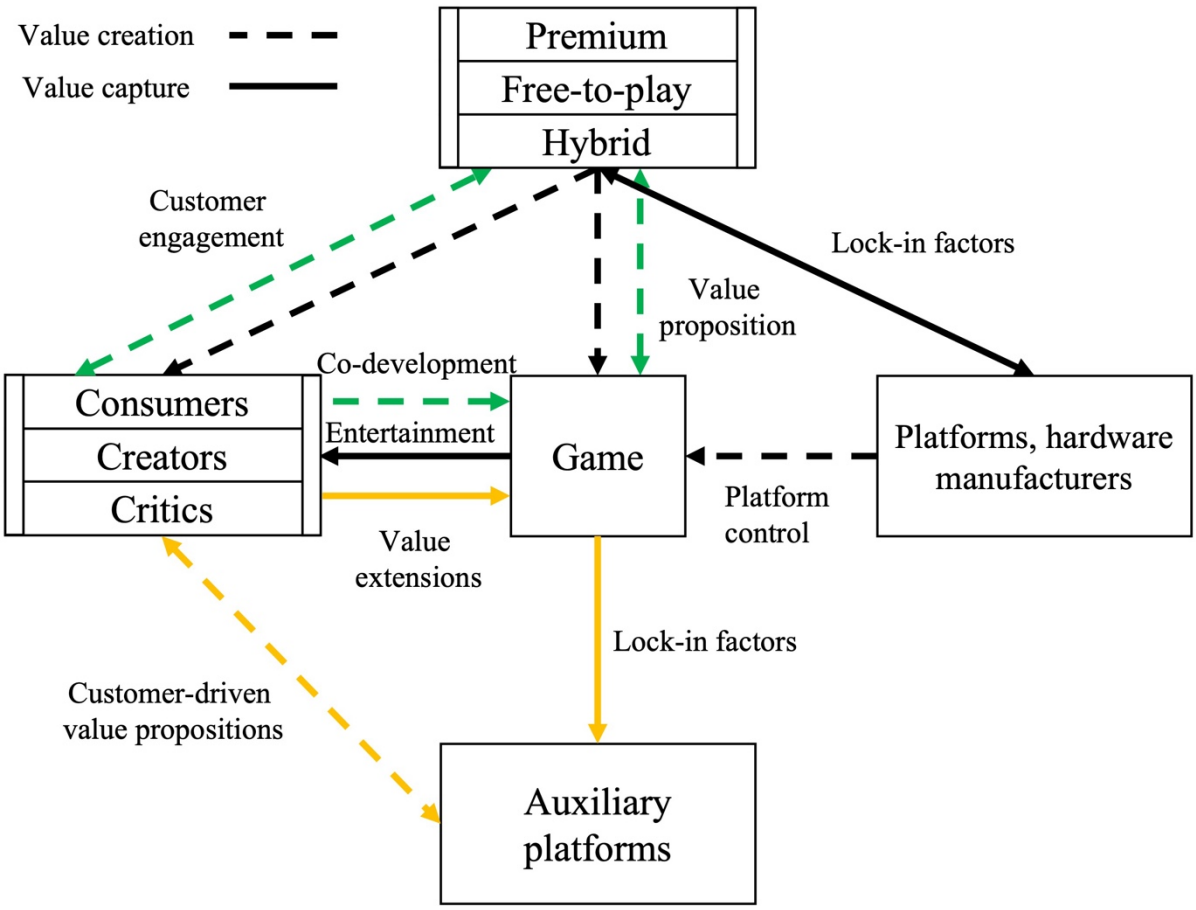


Figure 6. Value extension logics in the multiplex of value creation and capture in the video games industry.

Value is no longer tied to relationships where producers and platforms directly need to be involved, but instead consumers can interact with auxiliary platforms to create and capture value. Streaming video games and esports (e.g. Sjöblom & Hamari, 2017) are perhaps the most dominant examples in this perspective. By playing video games and utilizing streaming platforms, streamers generate revenue streams that are not necessarily connected to the firm that developed the game, and simultaneously consumers are experiencing the game without

actually playing it. While the games themselves and in-game items are predominantly purchased from the developers, in-game items can also be sold through secondary markets (auxiliary platforms) that are not necessarily connected to the game development company. In addition, consumers can utilize their own gameplay data to improve their skills in the game (learning aspect), professionally compete against other players, or share their gameplay on other platforms (e.g. Sjöblom & Hamari, 2017; Sjöblom et al., 2019). By the same token, the consumer can also become employed within the industry based on games related content they have created (a modding or streaming career) (e.g. Charezand & Geeroms, 2008). The reviewed literature, however, shows few cases of value extension where, for example, one firm creates value propositions that consumers utilize with other services to create meaningful and valuable experiences in their own context.

4.4 Value-in-use logics: Beyond traditional actors

While the previous three value creation and capture logics illustrate how value creation and capture have broadened and diversified between producers, consumers, and platforms, value-in-use draws attention to completely new value creation and capture logics where neither producers nor platforms are necessarily present anymore. Consumers now engage with games through various activities such as watching others play, creating or consuming fan art, and purchasing game-related merchandise. In this context, value capture for consumers extends beyond gameplay and becomes an experiential endeavor, where games serve as the foundation for diverse and innovative value propositions. While new media forms may be incorporated, the game remains the central starting point for value creation and capture. Blom (2021) has coined this approach as "ludo mix," a derivative of media mix. Although value creation and capture in this context may not always benefit producers and platforms, the underlying business logic focuses on leveraging the game as a controlled framework for

emerging value propositions and countering monetization attrition. In other words, by relinquishing immediate value capture efforts, companies emphasize capturing value through consumers' long-term commitment to the game or intellectual property.

Due to the proliferation of value creation and capture logics, studies falling under this category have diversified the research foci as the industry develops and the body of knowledge looking at the game industry has increased. More specifically, as business models and value propositions have significantly increased, prior research has looked at the ethical conditions and underpinnings of video game development (Bulut, 2015; Heimo et al., 2018) and similarly from a consumption point of view (Fantone, 2009; Haghshenas & Richards, 2016; Heeter et al., 2009; Kim & McClung, 2010; Schroeder & Borgerson, 2015). Figure 7 depicts different value creation and capture directions and areas under the value-in-use logic. Worth noting here is that the identified interaction between platforms and auxiliary platforms is a phenomenon that the extant literature does not seem to have covered, as we did not identify such research within the reviewed literature.

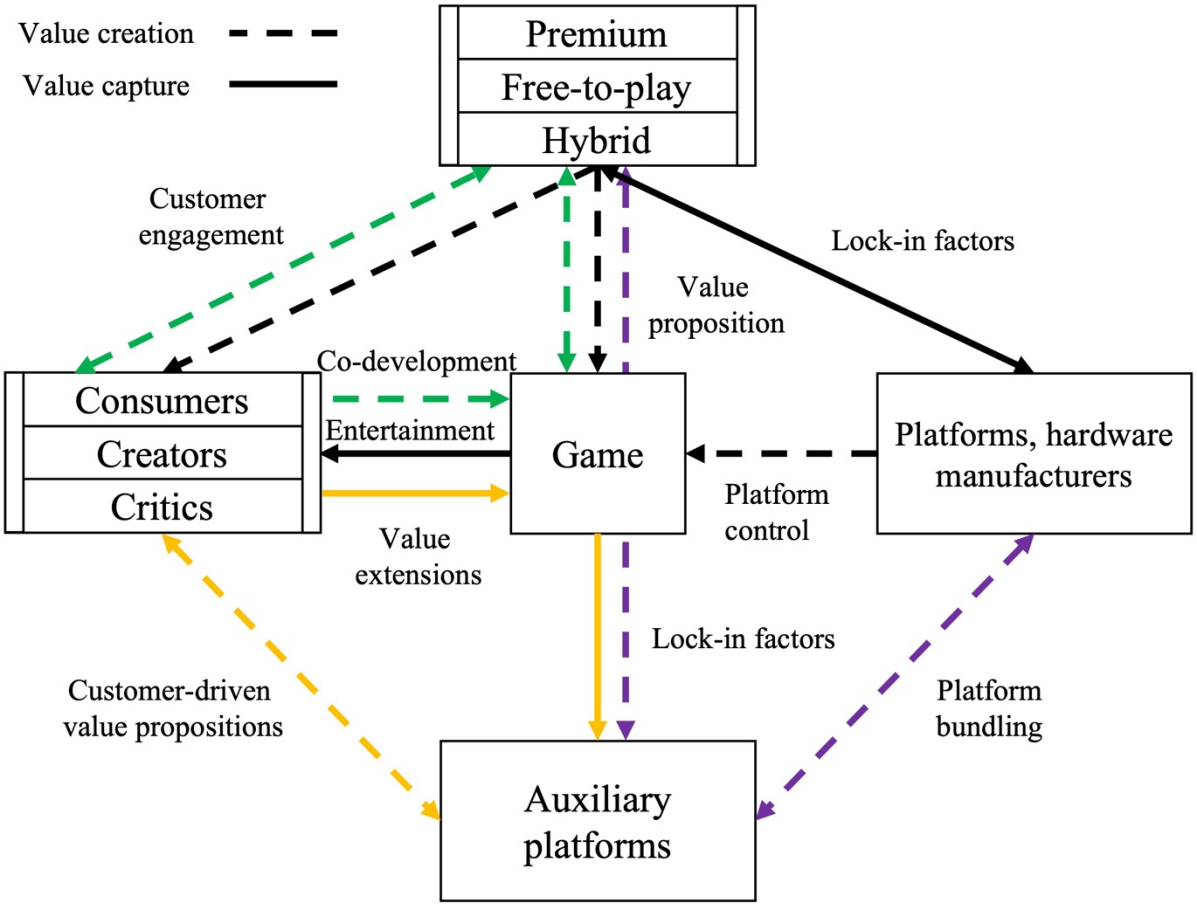


Figure 7. Value-in-use logics in the multiplex of value creation and capture in the video games industry.

The emergence of auxiliary platforms has been increasing steadily in the era of "value-in-use" (Parshakov et al., 2020). The kinds of value that consumers can create with and through games have extended beyond playing alone or with others, or even streaming gameplay. It now includes creating and influencing consumer behavior aspects and industries (Chess & Paul, 2019). Games, in the spirit of ludo mix, are used as the starting point for creating value propositions in other consumer goods domains such as clothing, food and beverages and other forms of media (Consalvo, 2009). For instance, game soundtracks are shared as musical artefacts to be listened to outside of games, and game spin-off movies and shows, based on games such as *The Witcher* Netflix series, are being produced. Games have become more

than the gaming activity itself (Denegri-Knott & Molesworth, 2010). They are cultural artefacts that derive their value, meaning, and arguably their success, from an increasing multiplex of value that can be created from and through them. Yet at the same time, video games and their inherent reliance on consumer participation seem to be at the core of any additional value proposition, for better or for worse. In other words, the game industry seems to have succeeded in devising novel value creation and capture logics because games engage the consumer in ways most other products and services currently do not.

Put differently, as technology provides consumers with new ways to engage with games and allows game development companies to create novel value propositions, we can expect to witness innovations in these areas. As Figure 7 reveals, we have identified value creation and capture domains (purple arrows) that the extant literature has so far left unexplored. Moreover, the relationship between auxiliary platforms and platforms distributing video games remains relatively unexplored in current body of knowledge, yet this research direction holds significant potential for contributing to platform studies. Essentially, platform bundling refers to value creation strategies that aim to capture platform value through lock-in factors, achieved through enforcement, acquisition, or cooperation. For example, the recent lawsuit between Apple and Epic Games (developer of Fortnite) (Porter, 2021) was a prime example of emerging platform wars that will most likely impact not only consumers but also the relationship between game development firms and various platforms. By the same token, in April 2023 the United Kingdom's Competition & Market Authority (2023) issued a report that suggested Microsoft's acquisition of Activision Blizzard should be blocked because the acquisition would significantly hinder cloud gaming competition in the future due to Microsoft's dominant position. Whether or not the acquisition goes through, what is worth

noting here is that content will become a key competitive factor for platforms when it comes to cloud gaming (i.e. games would no longer be played on dedicated hardware).

5. Discussion and future research avenues

5.1 Towards a multiplex of value creation and capture logics

Perhaps partly owing to rapid developments in the industry, our review has highlighted theoretical and conceptual shortcomings in prior literature in terms of how different value creation and capture logics operate and under what conditions. The video game industry is characterized by a co-existence of different value creation and capture logics that are interconnected in diverse ways. How the industry has developed and will continue to develop is not a simple matter of a new era taking over the existing one (Kerr, 2017) or new entrants disrupting the *status quo* (Zackariasson & Wilson, 2014).

Yet, at the same time we know how logics can simultaneously compete and co-exist on the field or industry level (e.g. Gawer & Phillips, 2013; Marquis & Lounsbury, 2007). Similar interdependencies are present in the video game industry as games enable scaffolding of value co-creation (Lin et al., 2015) and conversion of value (Malaby, 2006) in multiple ways across actors and platforms. For example, game development companies release games that consumers can either buy from a store (both physical or digital) or download and pay for according to their needs (i.e., free-to-play). To a large extent, this relatively straightforward model of value creation and capture is still prevalent in the video game industry, although digital means of distribution have become the dominant way of value creation and capture in video games. Technology has made it possible for game companies to actively monitor how

their games are played, allowing them to update or alter the game in light of how consumers are interacting with it, potentially creating a never-ending loop of value creation (Bauckhage et al., 2015). In essence, we refer to such simultaneous mobilizations of different value creation and capture logics as the multiplex of value creation and capture logics (Figure 8).

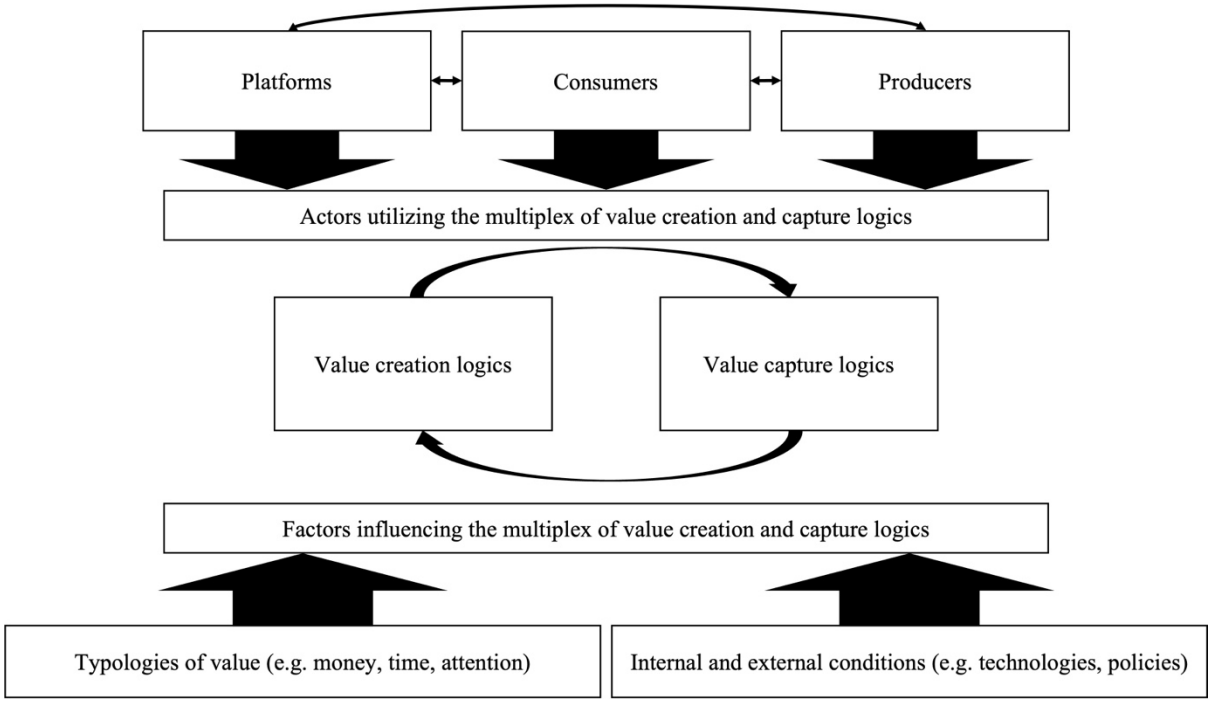


Figure 8. Integrative framework for the multiplex of value creation and capture logics.

While prior studies have predominantly focused on single value creation and capture logics, the integrative framework above suggests adopting a multiplex vantage point for making theoretical inroads on value creation and capture in digitized industries. Further, the framework also draws attention to the possibility that value creation and capture logics can be mobilized both simultaneously and sequentially. Similarly, this vantage point of the multiplex of value creation and capture logic presents a hitherto unexplored analytical environment to study value creation and capture. Further, this analytical environment is vastly interconnected and multidirectional, and dynamically evolving, akin to business and innovation ecosystems

but if that metaphor is conceptually accurate, remains to be investigated. The above, then, lends itself to novel and exciting research questions and avenues that we describe in more detail in the following section.

5.2 Future research avenues

Actors utilizing the multiplex of value creation and capture logics

Digitalization has impacted numerous industries and will continue to do so (e.g. Cenamor et al., 2017; Hagberg et al., 2016; Secundo et al., 2020) either by realigning or transforming intra- or interorganizational value chains or by blurring the boundaries between actors utilizing the multiplex of value creation and capture logics (e.g. platforms / intermediaries, producers and consumers). Under this light, the integrated model presented in this paper can provide a theoretically relevant lens for studying a diverse range of industries (e.g. retail, entertainment, academic entrepreneurship, and manufacturing) (Panourgias et al., 2014) well beyond the game industry. The speed with which the video game industry has been changing provides scholars with interesting opportunities to look at creativity and innovations development in the video game industry as a proxy for other creative industries similar in nature (O'Donnell, 2014; Zackariasson & Wilson, 2010, 2013, 2014). This fast moving and complex environment provides opportunities for multidisciplinary research of prosumers that could allow, for example, research to redefine who are consumers and producers in the digital age. How is creativity captured, monetized, or enacted upon, even if value creators are consumers rather than producers in the traditional meaning of the word? How does the blurring of lines between consumers and producers impact the traditional flow of value from producers to consumers and what does it mean for copyrights and payments of royalties for

example? What role do platforms and intermediaries play in this new ecosystem? How can they serve consumers as producers as much as they have traditionally taken upon themselves the roles of producers and developers? In terms of crucial theoretical inroads, we need research that asks: how do organizations mobilize different value creation and capture logics? Are such mobilizations simultaneous or sequential? In this regard, research on hybrid practices in institutional theory (e.g. Gautier et al., 2023) could provide a fruitful theoretical lens for investigating how firms blend together different institutional logics in hitherto unknown ways (Yan et al., 2021). In addition, future studies could utilize sensemaking approaches or multiple case studies (Langley, 2007) to shed light on institutional logics.

By the same token, and perhaps more prominently, the multiplex suggests rethinking our understanding of value as well as value creation and capture. Not only will there be differences between stakeholder groups in terms of what value is, (e.g. value discrepancies between consumers and producers / developers) but also the nature of games further complicate what value means to individuals within stakeholder groups (i.e. individual consumers, individual producers, separate platforms etc.,). For example, for some consumers, games are a means of escapism while for others, they are means for connecting with others, self-expression, education, or entertainment. Similarly, to developers, games for example can be a money-making business, or an artistic tool of self-expression. Platforms (e.g., Steam v. OnlyFans) differ in their definition of what creative value to capitalize on.

Hence, we need a more diversified understanding and examination of value in and of itself in different digitized, creative industries as well as what it means to different people within the same group of producers, consumers, and intermediaries, and how these differences impact value creation and capture. Put differently, we need to redefine what value is in the digital

age. If value is traditionally seen as something offered by producers and producers are no longer the sole creators of value, then what do we judge as value? According to whom? And under what conditions? To what extent can value creation and capture be described as direct / indirect, and how are they interrelated? As researchers, what types of value are we talking about when we conduct research on value creation and capture? From whose perspective is value defined? Here, one promising theoretical lens would focus on external search and especially from a coevolutionary lock-in perspective (Park et al., 2023) to shed light on how organizations' value propositions act as signals for customers and users on what kinds of value are favored over others. In addition, multimodal approaches to institutionalization and sensemaking (e.g. Meyer et al., 2018) hold considerable potential in analyzing how visual and verbal discourses construct meaning regarding value. In this regard, there is a need for studies that focus on multiple levels of analysis, utilize both primary and secondary data, as well as investigate value from a discursive or practice perspective.

Building on the above, in many industries, diversity, equity, and inclusion (DEI) initiatives are a trend, and they are so to a certain extent also in the video game industry. However, this is an industry where we see significant opposition especially from consumers towards DEI initiatives. Game accessibility is thought to "ruin" gameplay experiences as it is perceived to make games less challenging and, consequently, less engaging. Similarly, there are fears that DEI initiatives in companies would lead to more feminine games that would not have the same levels of challenge, violence, or gore that many consumers expect from games. How do accessibility to people with disabilities and DEI influence value creation and capture logics and vice versa? How, hence, do companies balance the needs of some consumers segments with the needs of other consumer segments? When demands and fears arise from a "true gamer" ideology, how are they to be navigated by the industry?

In the same vein of re-understanding value, as games have been transitioning from being products to becoming services, studies exploring the consumption of video games (e.g. bundling in the case of Derdenger & Kumar, 2013) could also be advanced by fine-tuning their conceptualization of value creation and capture when studying the video game industry. Further, while physical product bundles still exist, it is highly likely that physical cartridges will be phased out in the future or become a hobbyists' niche. As such, the evolving video game industry will continue to serve as an exciting context for theory-testing purposes of the evolving multiplex of value creation and capture. How independent organizations sustain and develop their competitive advantages? What are the implications of industry consolidation for coopetition and multi-level competition? What are the benefits of regional clustering in the era of platform wars?

Finally, given the theoretical and methodological plurality in the studies we have reviewed in this study, we believe such diversity should be emphasized also in future research. Having said that, above we have provided suggestions for more specific theoretical and methodological vantage points that we believe have the potential to generate impactful and thought-provoking findings. Furthermore, the integrative framework we presented above draws attention to the diversity of actors and the simultaneous presence of multiple, perhaps conflicting, logics, and such plurality should be reflected also in the chosen ontological and epistemological standpoint. What is of importance, here, is that future inquiries move away from seeing actors as well as value creation and capture as static entities.

Factors influencing the multiplex of value creation and capture logics

Another intriguing angle that warrants more research is that of the factors influencing the multiplex of value creation and capture logics (also by actors not identified in the integrative framework). More research is required regarding the extent of and through what mechanisms are the evolution and dynamics of the multiplex driven by one and / or multiple internal and external actors. Might there be other gravitational and / or external forces shaping the multiplex? Or, conversely, is it a question of capitalizing on *ad hoc* opportunities that can lead to the multiplex evolving with little or no steering? How do new value creation and capture logics emerge from novel technological innovations? Will new logics co-exist or compete with current ones? One potential avenue here would focus on adopting the configurational theorizing perspective (Furnari et al., 2021) given the relative complexity advocated in the integrative framework. In terms of methodologies, fuzzy set QCA and similar configurational methodologies (Fiss, 2011) can provide the necessary complexity for configurational theorizing. In addition, institutional theories, and institutional entrepreneurship (e.g. Zilber, 2007) in particular provides an interesting theoretical vantage point for exploring firm-level and field-level discourses and practices.

Furthermore, there is also a need for investigating phenomena within firms. For example, changes in revenue models, most prominently the proliferation of free-to-play initiated changes in the game design and development process (Marchand 2016), and also eventually in human resource management practices (who to recruit and what kind of skillsets to develop), marketing (marketing strategies during the game's lifecycle), and management (how to manage the design process and transition from one game towards new projects). Building on Heimo et al. (2018) and Bulut (2015) who studied game development and production from an ethical standpoint, more studies are needed to better understand how field level changes in practices have, in turn, effectively re-shaped the industry dynamics and how

firms adhere to, change, and / or resist such practices. As suggested by Anthony et al. (2023), ethnography would allow future inquiries to generate in-depth data on how game development and managerial practices are intertwined.

Extant research (Hsiao & Chen, 2016; Hui, 2017) has shed light on freemium or free-to-play mechanics in the video game industry, and this avenue holds tremendous potential for investigations that would not only contribute to our understanding of the game industry itself, but also of other industries where the freemium business model has been expanding, such as work-related communication tools (e.g. Slack), the music industry (e.g. Spotify), or survey and research tools (e.g. SurveyMonkey). For example, there is a paucity of studies looking at how consumers who do not spend any money on video games are converted into paying customers, as well as how companies conceptualize non-paying customers, and how a freemium model influences managerial decision-making in terms of recruitment matters and strategic directions. Indeed, while prior research has emphasized game development companies balance between creative freedom and profitability (Tschang, 2007) our integrative framework, coupled with recent developments in the industry, suggests tensions lie between profitability and ethical behavior (both game development and business models).

Future inquiries are encouraged to look into how video game development companies construct and make sense of value creation and capture and their perception of their moral, ethical, and environmental responsibility in such processes. We encourage researchers to investigate what role social responsibility, ethics, and morality play in defining, creating, capturing, and compensating value. Further, what kind of human resource practices should game companies employ to ensure the well-being of their employees and / or to cope with emerging changes such as artificial intelligence (AI) (e.g. impact on business, creativity, and

employment)? In this context, the video game industry can generate significant theoretical and empirical contributions on the impact AI has on organizations. Future studies could utilize Anthony et al.'s (2023) system view on AI or Kemp's (2023) concept of situated AI.

By the same token, how do commercial aspirations or technological advances influence game design aesthetics and practices? Is there a need to revisit and extend Tschang's (2007) prior theorization on balancing between profitability and creativity. We think so. As such, to what extent, and through what mechanisms, are companies allowed to capture value from their products? What kind of communicative strategies actors in the video game industry utilize to argue for or against ethical issues? In essence, more work is required inside game development companies as well as other relevant organizational actors to better understand how creativity, technologies, and business realities are intertwined. Anthony et al. (2023) advocated for ethnographic approaches, and we believe similar thick approaches are relevant also here. In addition, studies adhering to the material turn (Boxenbaum et al., 2018) would also be beneficial here in order to shed light on sociomaterial configurations that go beyond the cognitive domain.

Prior studies have often utilized the video game industry as a prime example of two-sided markets (e.g. Rysman, 2009). As our review has shown, the era of the video game industry as a relatively closed two-sided market might have come to an end mainly because physical distribution of video games is no longer the *de facto* means of selling games (Storz et al., 2015). However, it is possible that in the era of digital distribution, this two-sidedness status has shifted from being held by producers (Sony and Microsoft) to being held by digital distributers (Apple and Google). How this shift impacts the accentuation of various value creation logics within the multiplex of value creation and capture warrants further studies.

Furthermore, while the development of new gaming hardware is natural, it remains a costly endeavor especially when a firm has invested considerable resources in its ecosystem (e.g. Adner & Kapoor, 2010). As such, firms often attempt a balance between radical and incremental innovations through utilizing existing resources to develop new innovations, ensuring accumulated knowledge stocks are utilized also in the future.

In addition, due to digital distribution's rapid proliferation, regulation has not kept pace in many countries, thus creating a peculiar void where consumers are often left unprotected. For example, in the case of refunding digital purchases, video game development companies often direct the consumer to initiate a refund claim from the platform provider, and vice versa. Thus, what is required is both empirical and conceptual inquiries of who is responsible for what in this triadic relationship between the service provider, consumer, and platform? This poses questions of not just ethics as previously discussed, but also of regulation and customer protection (e.g. Dreier et al., 2017). In general, we need to ask: should creative industries / game industry be regulated? How? In what regards? And to the benefit of whom? How are new value creation and capture logics regulated and what are the strategic drivers behind organizational and industry-level self-regulation? As such, there is potential here for novel theoretical and empirical inroads through interdisciplinary research (e.g. management studies, sociology, political science, and / or moral philosophy).

5.3 Theoretical implications

We make two significant contributions. First, our integrated framework examines the multiplex of value creation and capture logics in the video game industry, providing a more detailed understanding of how value is created and captured. This framework recognizes the existence of multiple interconnected processes for creating and capturing value, which have

evolved over time due to socio-technical developments such as the introduction of new platforms, distribution models, and innovative approaches to value creation. These processes can occur with or without the involvement of a game development company. To gain a comprehensive perspective, it is crucial to move beyond traditional value creation and capture logics and consider their complex and intertwined nature. By doing so, we can identify the various forms of value that are established independently and collectively within this multiplex. Our framework acts as a valuable framework for analyzing and comprehending the evolution and current state of value creation within digitized industries at large.

The video game industry has been pioneering novel value creation and capture logics and processes by enabling consumers and developers to interact in new ways within and around games. As a result of advances in digital and platform business models, it can be assumed that multiplexity of value creation and capture will continue to evolve across industries.

Second, we assert unique evolutionary paths of multiplexity of value creation and capture for digitized industries. For instance, digital services are built more often than not by universally used tools (e.g. game engine Unity and programming languages Java and Swift), meaning that technological integrations across services can be created for enabling value creation and capture across industries and markets. Automotive industry is a prime example of value extension logics at play since some manufacturers have adopted microtransactions in enabling seat heating, smart device integration, and sharing data from vehicles with insurance companies. More specifically, technology provides impetus in both instances, but ultimately for the multiplexity to emerge, organizations are required to make strategic decisions on what boundaries to cross to create and capture value with and for their customers and stakeholders.

As such, we speculate that some industries are more lenient than others in terms of embracing the full spectrum of the value creation and capture multiplexity. Heavily regulated industries (e.g. banking, healthcare, automotive) might be more prone to adopting some of the value creation and capture logics, whilst industries less exposed to regulatory practices (e.g. media, retail, hospitality) could move towards embracing the multiplexity in its entirety.

This is in contrast to past studies on shifts from supply chains to ecosystems (Letaifa, 2014) or evolution of industry architecture (Jacobides, Knudsen, & Augier, 2006). Research that would study the evolution of multiplexity of value creation and capture across industries could compare and contrast various industries' value creation and capture dynamics and their evolutionary paths, and through that, better understand the general characteristics and differences of the evolution of value creation and capture multiplexity across industries. Our value creation and capture multiplexity model in the video game industry is also a solid vantage point for evaluating and envisioning future pathways for value creation and capture in digitized industries more broadly. We thus see much fruitful potential for future research on the multiplexity of value creation and capture to provide more accurate understanding of value creation and capture across industries and the various forms of value to those multiple actors involved in value creation and capture processes.

5.4 Managerial implications

While our integrative review is primarily intended to make novel theoretical contributions to the video game industry, as well as digitized industries more broadly, the integrative framework also holds significant implications for practitioners in and beyond the video game industry. More specifically, there are three key implications: creating and capturing value

from novel technologies, developing competences for navigating value creation and capture multiplexity, and competitive advantages through longevity and simplicity.

First, innovations emerging from the video game industry today (e.g. virtual reality, avatars) will likely be adopted in many other industries in the future, thus giving rise to similar kinds of evolutionary pathways in terms of value creation and capture logics. For instance, Netflix has started offering video games as part of its monthly subscription plans that had exclusively consisted of movies and TV series (Hersko, 2021). Further, real-time rendering, e.g. filters used in TikTok, was initially developed for video game development purposes (Failes, 2017), and has now become a part of most, if not all, social media applications. Thus, practitioners from outside the video game industry should observe how this industry develops and collaborate with video game development companies to generate first-mover advantages with novel technologies.

Second, managing and coordinating multiple value creation and capture logics points at developing new managerial and strategic competences in firms. For instance, many game development companies seemed to struggle with moving from premium to free-to-play business models, and as value creation and capture logics become more complex, game development companies need to acquire talent who can effectively navigate such complexities. What is of utmost importance here is that companies focus on developing such competences that enable them to create and capture value in an inclusive, accessible, and ethical fashion. Furthermore, one important aspect here focuses on creating a balanced set of value creation and capture logics within and across games to avoid alienating consumers while still allowing tie-ins and co-branding opportunities with companies from various

industries. This holds true especially for companies that use microtransactions and monthly subscriptions in their games.

Finally, amidst adopting novel technologies and managing the multiplexity of value creation and capture logics, game development companies should strive for competitive advantages through longevity and simplicity. For example, cryptocurrencies have already shown to be disastrous for the environment and similarly multiplexity can easily lead to user frustration through unnecessary complexity at the product / service level. Thus, longevity and simplicity are design and managerial decisions with strategic implications not only for the firm but also for the environment, users, and the society. In other words, while firms in the video game industry often compete with each other, these firms' impact extends well beyond the industry, which is why there is a need to strike a strategic balance between profitability and impact.

References

Abbasi, A. Z. and Jamak, A. B. S. A. (2017). 'Playful-Consumption Experience of Videogame-Play Influences Consumer Video-Game Engagement: A Conceptual Model'. *Global Business and Management Research*, **9**, 244-253.

Adner, R. and Kapoor, R. (2010). 'Value creation in innovation ecosystems: How the structure of technological interdependence affects firm performance in new technology generations'. *Strategic Management Journal*, **31**, 306-333.

Alpert, F. (2007). 'Entertainment software: suddenly huge, little understood'. *Asia Pacific Journal of Marketing and Logistics*, **19**, 87-100.

Anderson, E. G., Parker, G. G. and Tan, B. (2013). 'Platform Performance Investment in the Presence of Network Externalities'. *Information Systems Research*, **25**, 152-172.

Annosi, M. C., Ráez, R. M. O., Appio, F. P. and Del Giudice T. (2022). An integrative review of innovations in the agricultural sector: The roles of agency, structure, and their dynamic interplay. *Technological Forecasting and Social Change*, **185**, 122035.

Anthony, C., Bechky, B. A. and Fayard, A.-L. (2023). '"Collaborating" with AI: Taking a System View to Explore the Future of Work'. *Organization Science*, advance online publication. doi: 10.1287/orsc.2022.1651.

Aoyama, Y. and Izushi, H. (2003). 'Hardware gimmick or cultural innovation? Technological, cultural, and social foundations of the Japanese video game industry'. *Research Policy*, **32**, 423-444.

Arakji, R. Y. and Lang, K. R. (2007). 'Digital Consumer Networks and Producer-Consumer Collaboration: Innovation and Product Development in the Video Game Industry'. *Journal of Management Information Systems*, **24**, 195-219.

Arsenault, D. (2017). *Super Power, Spoony Bards, and Silverware: The Super Nintendo Entertainment System*. Cambridge, MA: MIT Press.

Autier, F. and Picq, T. (2005). ‘Is the resource-based 'view' a useful perspective for SHRM research? The case of the video game industry’. *International Journal of Technology Management*, **31**, 204-221.

Bagozzi, R. P. (1975). ‘Marketing as exchange’. *Journal of Marketing*, **39**, 32-39.

Balland, P-A., De Vaan, M. and Boschma, R. (2013). ‘The dynamics of interfirm networks along the industry life cycle: The case of the global video game industry, 1987-2007’. *Journal of Economic Geography*, **13**, 741-765.

Bauckhage, C., Drachen, A., & Sifa, R. (2015). Clustering Game Behavior Data. *IEEE Transactions on Computational Intelligence and AI in Games*, **7**, 266-278.

Binken, J. L. G and Stremersch, S. (2009). ‘The Effect of Superstar Software on Hardware Sales in System Markets’. *Journal of Marketing*, **73**, 88-104.

Blom, J. (2021). ‘Characters in Fire Emblem Three Houses: A Ludo Mix Perspective’. *Transactions of the Digital Games Research Association*, **5**, 101-130.

Boxenbaum, E., Jones, C., Meyer, R. E. and Svejenova, S. (2018). ‘Towards an articulation of the material and visual Turn in organization studies’. *Organization Studies*, **39**, 597-616.

Brandenburger, A. M. and Stuart Jr, H. W. (1996). ’Value-based business strategy’. *Journal of Economics & Management Strategy,* **5**, 5-24.

Brem, A., Bilgram, V. and Marchuk, A. (2019). 'How crowdfunding platforms change the nature of user innovation - from problem solving to entrepreneurship'. *Technological Forecasting & Social Change*, **144**, 348-360.

Broekhuizen, T. L. J., Lampel, J. and Rietveld, J. (2013). 'New horizons or a strategic mirage? Artist-led-distribution versus alliance strategy in the video game industry'. *Research Policy*, **42**, 954-964.

Browning, K. (2020). Chess (Yes, Chess) Is Now a Streaming Obsession. Available at: https://www.nytimes.com/2020/09/07/technology/chess-new-streaming-obsession-twitch.html (accessed 27 May 2023).

Bulut, E. (2015). 'Playboring in the Tester Pit: The Convergence of Precarity and the Degradation of Fun in Video Game Testing'. *Television & New Media*, **16**, 240-258.

Burger-Helmchen, T. and Cohendet, P. (2011). 'User Communities and Social Software in the Video Game Industry'. *Long Range Planning*, **44**, 317-343.

Bötel, J. (2019). Subnautica Postmortem. Available at: https://www.gdcvault.com/play/1025691/-Subnautica (accessed 27 May 2023).

Cabeza-Ramírez, L. J., Fuentes-García, F. J. and Muñoz-Fernandez, G. A. (2021). 'Exploring the Emerging Domain of Research on Video Game Live Streaming in Web of Science: State of the Art, Changes and Trends'. *International Journal of Environmental Research and Public Health*, **18**, 2917.

Cabras, I., Goumagias, N. D., Fernandes, K., Cowling, P., Li, F., Kudenko, D., Devlin, S. and Nucciarelli, A. (2017). 'Exploring survival rates of companies in the UK video-games industry: An empirical study'. *Technological Forecasting and Social Change*, **117**, 305-314.

Cadin, L. and Guérin, F. (2006). 'Guest Editorial'. *European Management Journal*, **24**, 247.

Cao, Y. and Downing, D. H. (2008). 'The realities of virtual play: video games and their industry in China'. *Media, Culture & Society*, **30**, 515-529.

Cenamor, J., Sjödin, D. R. and Parida, V. (2017). 'Adopting a platform approach in servitization: Leveraging the value of digitalization'. *International Journal of Production Economics*, **192**, 54-65.

Cenamor, J., Usero, B. and Fernandez, Z. (2013). 'The role of complementary products on platform adoption: Evidence from the video console market'. *Technovation*, **33**, 405-416.

Cennamo, C. and Santalo, J. (2013). 'Platform competition: Strategic trade-offs in platform markets'. *Strategic Management Journal*, **34**, 1331-1350.

Chatfield, T. (2011). *Fun inc: why games are the twenty-first century's most serious business*. Croydon: Virgin Books.

Chess, S. and Paul, C. A. (2019). 'The End of Casual: Long Live Casual'. *Games and Culture*, **14**, 107-118.

Christina, T. (2016). ‘The Implications of the Value Chain and Financial Institutions for Work and Employment: Insights from the Video Game Industry in Poland, Sweden and Germany’. *British Journal of Industrial Relations*, **54**, 311-333.

Clements, M. T. and Ohashi, H. (2005). ‘Indirect network effects and the product cycle: Video games in the US, 1994-2002’. *Journal of Industrial Economics*, **53**, 515-542.

Comunello, F. and Mulargia, S. (2015). ‘User-Generated Video Gaming: Little Big Planet and Participatory Cultures in Italy’. *Games and Culture*, **10**, 57-80.

Consalvo, M. (2006). ‘Console video games and global corporations: Creating a hybrid culture’. *New Media & Society*, **8**, 117-137.

Consalvo, M. (2009). ‘Convergence and Globalization in the Japanese Videogame Industry’. *Cinema Journal*, **48**, 135-141.

Copplestone, T. J. (2017). ‘But that’s not accurate: the differing perceptions of accuracy in cultural-heritage videogames between creators, consumers and critics’. *The Journal of Theory and Practice*, **21**, 415-438.

Cox, J. (2006). ‘Is there a first-mover advantage in the market for Japanese video game systems?’. *The Asia Pacific Journal of Economics & Business*, **10**, 18-33.

Cunningham, I., Loane, S. and Ibbotson, P. (2012). 'The internationalisation of small games development firms: evidence from Poland and Hungary'. *Journal of Small Business and Enterprise Development*, **19**, 246-262.

Darchen, S. and Tremblay, D-G. (2015). 'Policies for Creative Clusters: A Comparison between the Video Game Industries in Melbourne and Montreal'. *European Planning Studies*, **23**, 311-331.

Davidovici-Nora, M. (2014). 'Paid and Free Digital Business Models Innovations in the Video Game Industry'. *Communications & Strategies*, **94**, 83-102.

Denegri-Knott, J. and Molesworth, M. (2010). 'Concepts and practices of digital virtual consumption'. *Consumption Markets & Culture*, **13**, 109-132.

Derdenger, T. and Kumar, V. (2013). 'The Dynamic Effects of Bundling as a Product Strategy'. *Marketing Science*, **32**, 827-859.

Dixon, G. and Karboulonis, P. (2001). 'SMEs in European computer games and interactive entertainment software markets'. *International Journal of Entrepreneurship and Innovation Management*, **1**, 483-492.

Dreier, M., Wölfling, K., Duven, E., Giralt, S., Beutel, M. E. and Müller, K. W. (2017). 'Free-to-play: About addicted Whales, at risk Dolphins and healthy Minnows. Monetarization design and Internet Gaming Disorder'. *Addictive Behaviors*, **64**, 328-333.

Ebina, T. and Shimizu, D. (2009). 'Sequential mergers with differing differentiation levels'. *Australian Economic Papers*, **48**, 237-251.

Elsbach, K. D. and van Knippenberg, D. (2018). 'The *Academy of Management Annals*: Looking back, looking forward'. *Academy of Management Annals*, **12**, 1-4.

Elsbach, K. D. and van Knippenberg, D. (2020). 'Creating High-Impact Literature Reviews: An Argument for 'Integrative Reviews''. *Journal of Management Studies*, **57**, 1277-1289.

Ewalt, D. M. (2008). 'Nintendo's Power Up: How an oddly-named gizmo rebooted the video game industry'. *Harvard Asia Pacific Review*, **9**, 38-40.

Failes, I. (2017, 29 August). 'How Real-time Rendering Is Changing VFX And Animation Production'. *Cartoon Brew*. https://www.cartoonbrew.com/tools/real-time-rendering-changing-vfx-animation-production-153091.html

Fantone, L. (2009). 'Female players from margin to centre: female sociality, digital consumer citizenship and reterritorialisations'. *Digital Creativity*, **20**, 211-224.

Fiss, P. C. (2011). 'Building Better Causal Theories: A Fuzzy Set Approach to Typologies in Organization Research'. *Academy of Management Journal*, **54**, 393-420.

Furnari, S., Crilly, D., Misangyi, V. F., Greckhamer, T., Fiss, P. C. and Aguilera, R. V. (2021). 'Capturing Causal Complexity: Heuristics for Configurational Theorizing'. *Academy of Management Review*, **46**, 778-799.

Gallagher, S. and Park, S. H. (2002). 'Innovation and competition in standard-based industries: A historical analysis of the US home video game market'. *IEEE Transactions on Engineering Management*, **49**, 67-82.

Gandia, R. and Gardet, E. (2019). 'Sources of Dependence and Strategies to Innovate: Evidence from Video Game SMEs'. *Journal of Small Business Management*, **57**, 1136-1156.

Gautier, A., Pache, A.-C. and Santos, F. (2023). 'Making Sense of Hybrid Practices: The role of individual adherence to institutional logics in impact investing'. *Organization Studies*, advance online publication. doi: 10.1177/01708406231181693.

Gawer, A. (2014). 'Bridging differing perspectives on technological platforms: Toward an integrative framework'. *Research Policy*, **43**, 1239-1249.

Gawer, A. and Phillips, N. (2013). 'Institutional Work as Logics Shift: The Case of Intel's Transformation to Platform Leader'. *Organization Studies*, **34**, 1035-1071.

Gioia, D. A., Corley, K. G. and Hamilton, A. L. (2013). Seeking Qualitative Rigor in Inductive Research: Notes on the Gioia Methodology. *Organizational Research Methods*, **16**, 15-31.

Goggin, J. (2018). '"How Do Those Danish Bastards Sleep at Night?"*: Fan Labor and the Power of Cuteness'. *Games and Culture*, **13**, 747-764.

Greer, C. G. and Lei, D. (2012). 'Collaborative Innovation with Customers: A Review of the Literature and Suggestions for Future Research'. *International Journal of Management Reviews*, **12**, 63-84.

Gretz, R. T. (2010a). 'Console Price and Software Availability in the Home Video Game Industry'. *Atlantic Economic Journal*, **38**, 81-94.

Gretz, R. T. (2010b). 'Hardware quality vs. network size in the home video game industry'. *Journal of Economic Behavior & Organization*, **76**, 168-183.

Gretz, R. T. and Basuroy, S. (2013). 'Why Quality May Not Always Win: The Impact of Product Generation Life Cycles on Quality and Network Effects in High-tech Markets'. *Journal of Retailing*, **89**, 281-300.

Gretz, R. T., Malshe, A., Bauer, C. and Basuroy, S. (2019). 'The impact of superstar and non-superstar software on hardware sales: the moderating role of hardware lifecycle'. *Journal of the Academy of Marketing Science*, **47**, 394-416.

Grönroos, C. (2008). 'Service logic revisited: who creates value? And who co-creates?'. *European Business Review*, **20**, 298-314.

Grönroos, C. (2011). 'Value co-creation in service logic: A critical analysis'. *Marketing Theory*, **11**, 279-301.

Haefliger, S., Jager, P. and von Krogh, G. (2010). 'Under the radar: Industry entry by user entrepreneurs'. *Research Policy*, **39**, 1198-1213.

Hagberg, J., Sundstrom, M. and Egels-Zandén, N. (2016). 'The digitalization of retailing: an exploratory framework'. *International Journal of Retail & Distribution Management*, **44**, 694-712.

Haghshenas, H. and Richards, C. (2016). 'The Adoption of Mobile Game Applications as Sustainable Social and Behavioral Change'. *Competition Forum*, **14**, 108-118.

Hamari, J. and Sjöblom, M. (2017). 'What is eSports and why do people watch it?'. *Internet Research*, **27**, 211-232.

Hart, C. B. (Ed.) (2017). *The Evolution and Social Impact of Video Game Economics*. London: Lexington Books.

Harvey, J.-F., Cohendet, P., Simon, L. and Borzillo, S. (2015). 'Knowing Communities in the Front End of Innovation'. *Research-Technology Management*, **58**, 46-54.

Harviainen, J. T., Paavilainen, J. and Koskinen, E. (2020). 'Ayn Rand's Objectivist Ethics Applied to Video Game Business'. *Journal of Business Ethics*, **167**, 761-774.

Heeter, C., Egidio, R., Mishra, P., Winn, B. and Winn, J. (2009). 'Alien Games: Do Girls Prefer Games Designed by Girls?'. *Games and Culture*, **4**, 74-100.

Heimo, O. I., Harviainen, J. T., Kimppa, K. K. and Mäkilä, T. (2018). ’Virtual to Virtuous Money: A Virtue Ethics Perspective on Video Game Business Logic’. *Journal of Business Ethics*, **153**, 95-103.

Hersko, T. (2021, 20 July). ‘Netflix Posts Mixed Q2 Results, Will Soon Add Video Games for Subscribers’. *IndieWire*. https://www.indiewire.com/2021/07/netflix-q2-earnings-2021-video-games-subscriptions-1234652558/

Hodgson, D. and Briand, L. (2013). ‘Controlling the uncontrollable: 'Agile' teams and illusions of autonomy in creative work’. *Work, Employment and Society*, **27**, 308-325.

Hotho, S. and Champion, K. (2011). ‘Small businesses in the new creative industries: innovation as a people management challenge’. *Management Decision*, **49**, 29-54.

Hsiao, K-L. and Chen, C-C. (2016). ’What drives in-app purchase intention for mobile games? An examination of perceived values and loyalty’. *Electronic Commerce Research and Applications*, **16**, 18-29.

Huang, H. C. (2016). ’Freemium business model: construct development and measurement validation’. *Internet Research*, **26**, 604-625.

Hui, S-K. (2017). ’Understanding repeat playing behavior in casual games using a Bayesian data augmentation approach’. *Quantitative Marketing and Economics*, **15**, 29-55.

Huotari, P., Järvi, K., Kortelainen, S. and Huhtamäki, J. (2017). 'Winner does not take all: Selective attention and local bias in platform-based markets'. *Technological Forecasting and Social Change*, **114**, 313-326.

Inoue, Y. and Tsujimoto, M. (2018). 'New market development of platform ecosystems: A case study of the Nintendo Wii'. *Technological Forecasting and Social Change*, **136**, 235-253.

Izushi, H. and Aoyama, Y. (2006). 'Industry evolution and cross-sectoral skill transfers: a comparative analysis of the video game industry in Japan, the United States, and the United Kingdom'. *Environment and Planning A*, **38**, 1843-1861.

Jacobides, M. G., Knudsen, T., & Augier, M. (2006). Benefiting from innovation: Value creation, value appropriation and the role of industry architectures. *Research policy*, **35**(8), 1200-1221.

Jeppesen, L. B. and Molin, M. J. (2003). 'Consumers as co-developers: Learning and innovation outside the firm'. *Technology Analysis & Strategic Management*, **15**, 363-383.

Johns, J. (2006). 'Video games production networks: value capture, power relations and embeddedness'. *Journal of Economic Geography*, **6**, 151-180.

Johnson, M. R. and Woodcock, J. (2019). 'The impacts of live streaming and Twitch.tv on the video game industry'. *Media Culture & Society*, **41**, 670-688.

Johnson, R. S. (2013). ‘Toward Greater Production Diversity: Examining Social Boundaries at a Video Game Studio’. *Games and Culture*, **8**, 136-160.

Kang, S. J. and Kim, S. K. (2015). ‘Automated spatio-temporal analysis techniques for game environment’. *Multimedia Tools and Applications*, **74**, 6323-6329.

Kaur, P., Dhir, A., Tandon, A., Alzeiby, E. A. and Abohassan, A. A. (2021). ‘A systematic literature review on cyberstalking. An analysis of past achievements and future promises’. *Technological Forecasting & Social Change*, advance online publication. doi: 10.1016/j.techfore.2020.120426.

Kemp, A. (2023). ‘Competitive Advantages through Artificial Intelligence: Toward a Theory of Situated AI’. *Academy of Management Review*, advance online publication. doi: 10.5465/amr.2020.0205.

Kerr, A. (2006). *The Business and Culture of Digital Games: Gamework and Gameplay*. London: SAGE Publications.

Kerr, A. (2017). *Global Games: Production, Circulation and Policy in the Networked Era*. London: SAGE Publications.

Kim, M. S. and McClung, S. R. (2010). ‘Acceptability and ethics of product placement in sport video games’. *Journal of Promotion Management*, **16**, 411-427.

Koch, S. and Artmayr, P. (2020). 'Stability and development of user innovation strategies for video game producers'. *European Journal of Innovation Management*, **23**, 753-764.

Kumaraswamy, A., Garud, R. and Ansari, S. (2018). 'Perspectives on Disruptive Innovations'. *Journal of Management Studies*, **55**, 1025-1042.

Langley, A. (2007). 'Process thinking in strategic organization'. *Strategic Organization*, **5**, 271-282.

Lee, R. S. (2013). 'Vertical Integration and Exclusivity in Platform and Two-Sided Markets'. *The American Economic Review*, **103**, 2960-3000.

Lehtonen, M. J. and Harviainen, J. T. (2016). 'Mobile games and player communities: Designing for and with clans'. *Design Management Review*, **27**, 20-26.

Lehtonen, M. J., Vesa, M. and Harviainen, J. T. (2021). 'Games-as-a-Disservice: Emergent value co-destruction in platform business models'. *Journal of Business Research*, advance online publication. doi: 10.1016/j.jbusres.2021.11.055.

Lepak, D. P., Smith, K. G. and Taylor, M. S. (2007). 'Value creation and value capture: A multilevel perspective'. *Academy of Management Review*, **32**(1), 180-194.

Letaifa, S. B. (2014). The uneasy transition from supply chains to ecosystems: The value-creation/value-capture dilemma. *Management Decision*, **30**(10), 1115-1139

Liou, F. M. and Gao, Y. C. (2011). 'Competitive advantage in the online game industry in Taiwan'. *Journal of Strategy and Management*, **4**, 136-154.

Liu, H. J. (2010). 'Dynamics of Pricing in the Video Game Console Market: Skimming or Penetration?'. *Journal of Marketing Research*, **47**, 428-443.

Lusch, R. F. and Vargo, S. L. (2006). 'Service-dominant logic: reactions, reflections and refinements'. *Marketing Theory*, **6**, 281-8.

Marchand, A. (2016). 'The power of an installed base to combat lifecycle decline: The case of video games'. *International Journal of Research in Marketing*, **33**, 140-154.

Marchand, A. and Hennig-Thurau, T. (2013). 'Value Creation in the Video Game Industry: Industry Economics, Consumer Benefits, and Research Opportunities'. *Journal of Interactive Marketing*, **27**, 141-157.

Markman, G. D., Lieberman, M., Leiblein, M., Wei, L. Q. and Wang, Y. (2021). 'The Distinctive Domain of the Sharing Economy: Definitions, Value Creation, and Implications for Research'. *Journal of Management Studies*, **58**, 927-948.

Marquis, C. and Lounsbury, M. (2007). 'Vive la résistance: Competing logics and the consolidation of U.S. community banking'. *Academy of Management Journal*, **50**, 799–820.

Martin, C. B. and Deuze, M. (2009). 'The Independent Production of Culture: A Digital Games Case Study'. *Games and Culture*, **4**, 276-295.

Maruyama, M. and Ohkita, K. (2011). 'Platform Strategy of Video Game Software in Japan, 1984-1994: Theory and Evidence'. *Managerial and Decision Economics*, **32**, 105-118.

McIntyre, D. P. and Srinivasan, A. (2017). 'Networks, platforms, and strategy: Emerging views and next steps'. *Strategic Management Journal*, **38**, 141-160.

Meyer, R. E., Jancsary, D., Höllerer, M. A. and Boxenbaum, E. (2018). 'The Role of Verbal and Visual Text in the Process of Institutionalization'. *Academy of Management Review*, **43**, 392-418.

Molinillo, S., Japutra, A. and Liebana-Cabanillas, F. (2020). 'Impact of perceived value on casual mobile game loyalty: The moderating effect of intensity of playing'. *Journal of Consumer Behaviour*, **19**, 493-504.

Moran, P. and Ghoshal, S. (1996). 'Value creation by firms'. In J. B. Keys and L. N. Dosier (Eds.), *Academy of Management Best Paper Proceedings*, 41-45.

Möller, K. (2006). 'Role of competences in creating customer value: A value-creation logic approach'. *Industrial Marketing Management*, **35**, 913-924.

Nair, H. (2007). 'Intertemporal price discrimination with forward-looking consumers: Application to the US market for console video-games'. *Quantitative Marketing and Economics*, **5**, 237-292.

Nucciarelli, A., Li, F., Fernandes, K. J., Goumagias, N., Cabras, I., Devlin, S., Kudenko, D. and Cowling, P. (2017). 'From value chains to technological platforms: The effects of crowdfunding in the digital game industry'. *Journal of Business Research*, **78**, 341-352.

O'Donnell, C. (2014). *Developer's Dilemma: The Secret World of Videogame Creators*. Cambridge, MA: MIT Press.

Oh, Y. K. and Min, J. (2015). 'The Mediating Role of Popularity Rank on the Relationship between Advertising and In-app Purchase Sales in Mobile Application Market'. *Journal of Applied Business Research*, **31**, 1311-1322.

Panourgias, N. S., Nandhakumar, J. and Scarbrough, H. (2014). 'Entanglements of creative agency and digital technology: A sociomaterial study of computer game development'. *Technological Forecasting & Social Change*, **83**, 111-126.

Park, S., Piezunka, H. and Dahlander, L. (2023). 'Coevolutionary Lock-In in External Search'. *Academy of Management Journal*, advance online publication. doi: 10.5465/amj.2022.0710.

Parker, O. N., Krause, R. and Covin, J. G. (2017). 'Ready, Set, Slow: How Aspiration-Relative Product Quality Impacts the Rate of New Product Introduction'. *Journal of Management*, **43**, 2333-2356.

Parmentier, G. and Gandia, R. (2013). 'Managing Sustainable Innovation with a User Community Toolkit: The Case of the Video Game Trackmania'. *Creativity and Innovation Management*, **22**, 195-208.

Parmentier, G. and Mangematin, V. (2014). 'Orchestrating innovation with user communities in the creative industries'. *Technological Forecasting & Social Change*, **83**, 40-53.

Parry, G., Bustinza, O. F. and Vendrell-Herrero, F. (2012). 'Servitisation and value co-production in the UK music industry: An empirical study of Consumer Attitudes'. *International Journal of Production Economics*, **135**, 320-332.

Parshakov, P., Naidenova, I. and Barajas, A. (2020). 'Spillover effect in promotion: Evidence from video game publishers and eSports tournaments'. *Journal of Business Research*, **118**, 262-270.

Peltoniemi, M. (2015). 'Cultural Industries: Product–Market Characteristics, Management Challenges and Industry Dynamics'. *International Journal of Management Reviews*, **17**, 41-68.

Peng, H., Sun, Y-I. and Wu, H. (2009). 'Analysis of subscription characteristics and user mobility in different types of MMORPGs'. *Journal of China Universities of Posts and Telecommunications*, **16**, 121-128.

Pera, R., Menozzi, A., Abrate, G. and Baima, G. (2021). 'When cocreation turns into codestruction'. *Journal of Business Research*, **128**, 222-232.

Perks, H., Kowalkowski, C., Witell, L. and Gustafsson, A. (2017). 'Network orchestration for value platform development'. *Industrial Marketing Management*, **67**, 106-121.

Peticca-Harris, A., Weststar, J. and McKenna, S. (2015). 'The perils of project-based work: Attempting resistance to extreme work practices in video game development'. *Organization*, **22**, 570-587.

Porter, J. (2021). Epic Games brings Apple fight to the EU with new antitrust complaint. Available at: https://www.theverge.com/2021/2/17/22286998/epic-games-apple-european-comission-antitrust-complaint-app-store-fortnite (accessed 7 March 2021).

Postigo, H. (2007). 'Of mods and modders: Chasing down the value of fan-based digital game modifications'. *Games and Culture*, **2**, 300-313.

Prahalad, C. K. and Ramaswamy, V. (2004). 'Co-creation experiences: The next practice in value creation'. *Journal of Interactive Marketing*, **18**, 5-14.

Prieger, J. E. and Hu, W. M. (2012). 'Applications barrier to entry and exclusive vertical contracts in platform markets'. *Economic Inquiry*, **50**, 435-452.

Quandt, T., Van Looy, J., Vogelgesang, J., Elson, M., Ivory, J. D., Consalvo, M. and Mäyrä, F. (2015). 'Digital Games Research: A Survey Study on an Emerging Field and Its Prevalent Debates'. *Journal of Communication*, **65**, 975-996.

Quinn, J. B., Doorley, T. L. and Paquette, P. C. (1990). 'Beyond products: services-based strategy'. *Harvard Business Review*, **68**, 58-60.

Rajala, R., Rossi, M., Tuunainen, V. and Vihinen, J. (2007). 'Revenue Logics of Mobile Entertainment Software - Observations from Companies Producing Mobile Games'. *Journal of Theoretical and Applied Electronic Commerce Research*, **2**, 34-47.

Ramaswamy, V. (2008). 'Co-creating value through customers' experiences: the Nike case'. *Strategy & Leadership*, **36**, 9-14.

Ravasi, D. and Stigliani, I. (2012). 'Product design: A review and research agenda for management studies'. *International Journal of Management Reviews*, **14**, 464-488.

Rayna, T. and Striukova, L. (2014). ''Few to Many': Change of Business Model Paradigm in the Video Game Industry'. *Communications & Strategy*, **94**, 61-81.

Readman, J. and Grantham, A. (2006). 'Shopping for Buyers of Product Development Expertise: How Video Games Developers Stay Ahead'. *European Management Journal*, **24**, 256-269.

Rietveld, J., Schilling, M. A. and Bellavitis, C. (2019). 'Platform Strategy: Managing Ecosystem Value Through Selective Promotion of Complements'. *Organization Science*, **30**, 1232-1251.

Roberts, D., Hughes, M. and Kertbo, K. (2015). 'Exploring consumers' motivations to engage in innovation through co-creation activities'. *European Journal of Marketing*, **48**, 147-169.

Rysman, M. (2009). 'The Economics of Two-Sided Markets'. *Journal of Economic Perspectives*, **23**, 125-143.

Schilling, M. A. (2003). 'Technological leapfrogging: Lessons from the US video game console industry'. *California Management Review*, **45**, 6-32.

Schroeder, J. E. and Borgerson, J. L. (2015). 'Critical visual analysis of gender: reactions and reflections'. *Journal of Marketing Management*, **31**, 1723-1731.

Secundo, G., Rippa, P. and Cerchione, R. (2020). 'Digital Academic Entrepreneurship: A structured literature review and avenue for a research agenda'. *Technological Forecasting & Social Change*, advance online publication. doi: 10.1016/j.techfore.2020.120118.

Seidl, A., Caulkins, J. P., Hartl, R. F. and Kort, P. M. (2017). 'Serious strategy for the makers of fun: Analyzing the option to switch from pay-to-play to free-to-play in a two-stage optimal control model with quadratic costs'. *European Journal of Operational Research*, **267**, 700-715.

Shafer, S. M., Smith, H. J. and Linder, J. C. (2005). 'The power of business models'. *Business Horizons*, **48**, 199-207.

Shankar, V. and Bayus, B. L. (2003). 'Network effects and competition: An empirical analysis of the home video game industry'. *Strategic Management Journal*, **24**, 375-384.

Sjöblom, M. and Hamari, J. (2017). 'Why do people watch others play video games? An empirical study on the motivations of Twitch users'. *Computers in Human Behavior*, **75**, 985-996.

Sjöblom, M., Törhönen, M., Hamari, J., Macey, J. (2019). 'The ingredients of Twitch streaming: Affordances of game streams'. *Computers in Human Behavior*, **92**, 20-28.

Smith, A. N. (2015). 'The backer-developer connection: Exploring crowdfunding's influence on video game production'. *New Media & Society*, **17**, 198-214.

Snyder, H. (2019). Literature review as a research methodology: An overview and guidelines. *Journal of Business Research*, **104**, 333-339.

Storz, C., Riboldazzi, F. and John, M. (2015). 'Mobility and innovation: A cross-country comparison in the video games industry'. *Research Policy*, **44**, 121-137.

Subramanian, A. M., Chai, K. H. and Mu, S. (2011). 'Capability reconfiguration of incumbent firms: Nintendo in the video game industry'. *Technovation*, **31**, 228-239.

Sun, B., Xie, J. and Cao, H. H. (2004). 'Product Strategy for Innovators in Markets with Network Effects'. *Marketing Science*, **23**, 243-254.

Takeaki, W. (2011). ‘Exploitation Reduces Novelty: An Empirical Analysis of the Japanese Video Game Industry’. *Annals of Business Administrative Science*, **10**, 1-12.

Tang, A. K. Y. (2017). ‘Key factors in the triumph of Pokémon GO’. *Business Horizons*, **60**, 725-728.

Toffler, A. (1980). *The Third Wave*. New York: William Morrow and Company.

Torraco, R. J. (2005). Writing Integrative Literature Reviews: Guidelines and Examples. *Human Resource Development Review*, **4**, 356-367.

Tranfield, D., Denyer, D. and Smart, P. (2003). ‘Towards a methodology for developing evidence-informed management knowledge by means of systematic review’. *British Journal of Management*, **14**, 207-22.

Tschang, F. T. (2007). ‘Balancing the Tensions Between Rationalization and Creativity in the Video Games Industry’. *Organization Science*, **18**, 989-1005.

Törhönen, M., Giertz, J., Weiger, W. H. and Hamari, J. (2021). ‘Streamers: the new wave of digital entrepreneurship? Extant corpus and research agenda’. *Electronic Commerce Research and Applications*, **46**. doi: 10.1016/j.elerap.2020.101027

Törhönen, M., Sjöblom, M., Hassan, L. and Hamari, J. (2020). ’Fame and Fortune, or Just Fun? A study on why people create content on video platforms’. *Internet Research*, **30**, 165-190.

Valor, C., Antonetti, P. and Crisafulli, B. (2022). Emotions and consumers' adoption of innovations: An integrative review and research agenda. *Technological Forecasting and Social Change*, **179**, 121609.

van den Ende, J., Jaspers, F. and Rijsdijk, S. A. (2013). 'Should System Firms Develop Complementary Products? A Dynamic Model and an Empirical Test'. *Journal of Product and Innovation Management*, **30**, 1178-1198.

Vandermerwe, S. and Rada, J. (1988). 'Servitization of Business: Adding Value by Adding Services'. *European Management Journal*, **6**, 314-324.

Vargo, S. L., Lusch, R. F. (2004). 'Evolving to a New Dominant Logic for Marketing'. *Journal of Marketing*, **68**, 1-17.

Vargo, S. L., Lusch, R. F. (2008). 'Service-dominant logic: continuing the evolution'. *Journal of the Academy of Marketing Science*, **36**, 1-10.

Venkatraman, N. and Lee, C. H. (2004). 'Preferential linkage and network evolution: A conceptual model and empirical test in the US video game sector'. *Academy of Management Journal*, **47**, 876-892.

Waldner, F., Zsifkovits, M. and Heidenberger, K. (2013). 'Are service-based business models of the video game industry blueprints for the music industry?'. *International Journal of Services, Economics and Management*, **5**, 5-20.

Walfisz, M., Zackariasson, P. and Wilson, T. L. (2006). 'Real-time strategy: Evolutionary game development'. *Business Horizons*, **49**, 487-498.

Walters, D. and Lancaster, G. (2000). 'Implementing value strategy through the value chain'. *Management Decision*, **38**, 160-178.

Wang, Q. H., Mayer-Schönberger, V. and Yang, X. (2013). 'The determinants of monetary value of virtual goods: An empirical study for a cross-section of MMORPGs'. *Information Systems Frontiers*, **15**, 481-495.

Wang, T., Wu, D., Zhang, J., Chen, M. and Zhou, Y. (2016). 'Measuring and Analyzing Third-Party Mobile Game App Stores in China'. *IEEE Transactions on Network and Service Management*, **13**, 793-805.

Whittle, A., Vaara, E. and Maitlis, S. (2023). The Role of Language in Organizational Sensemaking: An Integrative Theoretical Framework and an Agenda for Future Research. *Journal of Management*, advance online publication. doi: 1 0.1177/01492063221147295.

Wieland, H., Koskela-Huotari, K. and Vargo, S. L. (2016). 'Extending actor participation in value creation: an institutional view'. *Journal of Strategic Marketing*, **24**, 210-226.

Wilson, T. L and Zackariasson, P. (2012). 'Through the Looking Glass Sharply'. In Zackariasson, P. and Wilson, T. L. (Eds), *The Video Game Industry*. London: Routledge, 244-262.

Xu, F., Buhalis, D. and Weber, J. (2017). 'Serious games and the gamification of tourism'. *Tourism Management*, **60**, 244-256.

Xu, X-Y., Luo, X., Wu, K. and Zhao, W-J. (2021). 'Exploring viewer participation in online video game streaming: A mixed-methods approach'. *International Journal of Information Management*, **58**, 102297.

Yan, S., Almandoz, J. and Ferraro, F. (2021). 'The Impact of Logic (In)Compatibility: Green Investing, State Policy, and Corporate Environmental Performance'. *Administrative Science Quarterly*, **66**, 903-944.

Yoo, Y., Boland Jr, R. J., Lyytinen, K. and Majchrzak, A. (2012). 'Organizing for innovation in the digitized world'. *Organization Science*, **23**, 1398-1408.

Zackariasson, P. and Wilson, T. (2010). 'Paradigm shifts in the video game industry'. *Competitiveness Review*, **20**, 139-151.

Zackariasson, P. and Wilson, T. (2013). 'The New Business Logics of Video Games: Triple Evolutionary Processes in Perspective'. *Competition Forum*, **11**, 56-64.

Zackariasson, P. and Wilson, T. (Eds) (2014). *The Video Game Industry: Formation, Present State, and Future*. London: Routledge.

Zhou, Y. Y. (2017). ‘Bayesian Estimation of a Dynamic Model of Two-Sided Markets: Application to the US Video Game Industry’. *Management Science*, **63**, 3874-3894.

Zhu, F. and Iansiti, M. (2012). ‘Entry into platform-based markets’. *Strategic Management Journal*, **33**, 88-106.

Zilber, T. B. (2007). ‘Stories and the Discursive Dynamics of Institutional Entrepreneurship: The Case of Israeli High-tech after the Bubble’. *Organization Studies*, **28**, 1035-1054.

Zimmerman, J. J. (2019). ‘Computer Game Fan Communities, Community Management, and Structures of Membership’. *Games & Culture*, **14**, 896-916.